

Chinese-SkillSpan: A Versioned Resource for Competency Span Extraction from Chinese Job Advertisements

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ABSTRACT

Background. Comparing competency extraction from Chinese job advertisements requires explicit rules for coordinated phrases, implicit word boundaries, and context-dependent types. Existing resources leave these decisions unsettled for naturally occurring Chinese text.

Methods. We developed Chinese-SkillSpan from 22,840 sentences across four recruitment collections. An ESCO-informed handbook defines character spans for language, knowledge, occupational skills, and transversal competences. LLM-generated supervision and a fixed 150-sentence human reference support model comparison; the reference also informed guideline development. Three annotators independently coded a separate random sample of 50 sentences, blinded to machine suggestions and one another’s labels. We evaluated prompted models, supervised adaptation, and multilingual encoder baselines.

Results. Qwen adaptation improved extraction across three training seeds under fixed inference rules. The Chinese encoder comparison did not establish an advantage of ESCO initialization. Independent blinded coding showed greater agreement on boundaries alone than on complete typed spans: mean typed exact F1 was 0.663 (95% CI [0.562, 0.759]). Both human and model assessments therefore identify boundary and type decisions as remaining sources of disagreement.

Conclusions. Chinese-SkillSpan makes annotation decisions inspectable and supports diagnostic comparisons of models and supervision. Its separate coding study identifies unresolved annotation ambiguity. Generalization beyond the development-used human reference remains to be established.

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INTRODUCTION

Online job advertisements describe the knowledge and skills employers seek and provide evidence of changing occupational requirements (Podjanin and Strietska-Ilina, 2020; Senger et al., 2024). Skill-demand analysis and job–curriculum matching require both the requested competency and the words used to express it (Bhola et al., 2020; Gnehm et al., 2022). Job Skill Named Entity Recognition (JobSkillNER) identifies these mentions and assigns a type to each span. Keeping the source wording allows readers to inspect decisions that a document-level skill inventory would conceal.

For Chinese advertisements, even a short phrase can require several annotation decisions. A technology name can denote tool use in one clause and knowledge of principles in another. Coordination may express separate activities or share an action that must remain attached to its objects. Chinese text normally lacks spaces between words, so tokenization alone cannot resolve these semantic boundaries (Peng and Dredze, 2015; Chen and Kong, 2021). Similar ambiguities occur in other languages (Nguyen et al., 2024), but translating their datasets does not determine which Chinese substrings to select. The task calls for explicit, context-sensitive rules; Table 3 pairs Chinese examples with English explanations.

Existing resources provide precedents, but their annotation schemes differ. English SkillSpan permits overlapping SKILL and KNOWLEDGE layers (Zhang et al., 2022b). Kompetenzer adds ESCO-derived fine-grained classification to Danish and English spans (Zhang et al., 2022a). German recruitment work extracts education, experience, and language requirements before finer classification (Gnehm et al., 2022). Document-level retrieval and Chinese skill-demand forecasting use other prediction targets (Bhola et al., 2020; Chen et al., 2024).

We introduce Chinese-SkillSpan to study these decisions. It combines a corpus of 22,840 sentences from four recruitment collections with a handbook for language (L), knowledge (K), occupational skills (S), and transversal competences (T). The scheme follows ESCO’s high-level categories and the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017), without assigning ESCO concept identifiers. Character offsets preserve the wording of each mention.

The study asks three questions. How should Chinese mentions be delimited and typed when coordination, omitted predicates, and context defeat a word-based inventory? The handbook provides source-contiguous spans and positive and negative cases. What evidence supports the annotations? Independent blinded coding by three annotators is assessed separately from historical coding and assisted Silver review, with each layer’s versions retained. How do models behave under these conventions? Matched Qwen adaptation, Chinese-supervised encoders, and boundary/type diagnostics examine performance on the frozen human reference.

These contributions connect task definition, annotation evidence, and model assessment. The original corpus establishes the resource’s scope; it is not interchangeable with the smaller labeled training and evaluation subsets. The 150-sentence human reference contributed to guideline development and supports diagnostic assessment rather than an independent leaderboard. External recruitment experiments place the encoder comparison in context without treating scores from different languages and label schemes as a shared ranking. We document the role and availability of each subset following guidance on dataset documentation and annotation agreement (Bender and Friedman, 2018; Gebru et al., 2021; Artstein and Poesio, 2008).

RELATED WORK

Skill extraction and recruitment datasets. Recruitment datasets differ first in what a system must predict. Sayfullina et al. (2018) ask whether a candidate soft-skill phrase refers to an applicant, whereas Bhola et al. (2020) retrieve skills from a predefined inventory for an entire advertisement. Span extraction locates the words expressing a requirement. Within that task, SkillSpan annotates English SKILL and KNOWLEDGE in two potentially overlapping layers; hard and soft skills describe its coverage rather than its label inventory (Zhang et al., 2022b). Kompetenzer combines manually identified skill and knowledge spans with ESCO-derived fine-grained labels in Danish and English, with manual correction of the distantly supervised Danish test labels (Zhang et al., 2022a). Gnehm et al. (2022) extract coarse EDU/EXP/LNG spans before refining skill areas for taxonomy matching.

Chinese recruitment research includes both forecasting and entity extraction. Job-SDF evaluates skill-demand forecasting at several aggregation levels (Chen et al., 2024). For occupational profiling, Cao et al. (2021) manually annotated skill and specialty requirements using BIOES and trained a BERT-BiLSTM-CRF model. Chinese-SkillSpan instead uses flat character spans with four types and explicit rules for coordination and contextual interpretation. Its length statistics count Chinese source characters, so they cannot be compared directly with SkillSpan's token counts.

Taxonomy-based weak supervision can reduce manual annotation work (Zhang et al., 2022c), but generated labels still need validation. Nguyen et al. (2024) report difficulties with conjoined and ambiguous mentions across six recruitment datasets. Senger et al. (2024) also identify scarce public annotations and inconsistent terminology. Table 1 compares the prediction units, annotation layers, and available materials of selected resources.

Chinese span annotation and competency guidelines. Chinese NER benchmarks address social media and fine-grained entity typing (Peng and Dredze, 2015; Xu et al., 2020), while multilingual benchmarks include noisy inputs and contextual type distinctions (Fetahu et al., 2023; Luo et al., 2025). The absence of explicit word delimiters makes boundary decisions especially consequential for Chinese extraction (Chen and Kong, 2021). Domain-specific guidance can address additional problems: Zhong et al. (2023), for example, develop a rehabilitation-medicine corpus and investigate long and nested entities with BERT-Span. In recruitment text, annotators must also decide whether an expression denotes knowledge, an occupational action, or a transversal competence in its particular clause.

ESCO provides a multilingual vocabulary for labor-market information (le Vrang et al., 2014; European Commission, 2024). SkillLens combines human involvement, skill extraction, and ESCO linking across European job advertisements (De Santo et al., 2026). Chinese-SkillSpan adapts ESCO's high-level categories to LSKT annotation in Chinese. Its contribution is the operational guidance needed to select complete spans and type them in context. Agreement analysis and adjudication consequently address both unitization and categorization (Artstein and Poesio, 2008).

Model-assisted annotation and extraction models. LLMs can supply first-pass labels, with their usefulness depending on task instructions and human validation (Ding et al., 2023; Li et al., 2023; Ul Haq et al., 2026). GoLLIE and subsequent guideline-refinement work study how annotation instructions shape model outputs (Sainz et al., 2024; Kim et al., 2026). In recruitment, JobScape generates synthetic advertisements for skill-to-taxonomy matching (Magron et al., 2024).

Table 1. Selected recruitment extraction resources. Annotation layers and prediction units differ; corpus-scale counts are not compared with human-annotated sample sizes. Material descriptions refer to the cited work, not an exhaustive audit of every later release.

Work / language	Prediction unit	Actual annotation layers	Material scope
Sayfullina et al. (2018); English	Candidate phrase in context	Relevance to an applicant; binary disambiguation, not open-boundary extraction	Annotated soft-skill contexts
Gnehm et al. (2022); German	Spans, then finer areas	Coarse EDU/EXP/LNG; finer skill areas and taxonomy matching	Article and model; full annotated corpus access not established here
Zhang et al. (2022b); English	Source-text spans	SKILL / KNOWLEDGE; overlap permitted between the two layers	Dataset, code, and annotation guidance
Zhang et al. (2022a); Danish / English	Spans and fine-grained labels	Manual skill/knowledge spans; ESCO API distant labels; Danish test-label correction	Dataset and code
Cao et al. (2021); Chinese	Recruitment entities	Manually annotated skill / specialty entities; BIOES	Article and supporting data; full NER corpus access not established here
This work; Chinese	Flat character spans	L/K/S/T; 9,540 Silver records; 500 human-coded/reviewed sentences, including Gold150	Version-specific reference, prompts, predictions and scoring materials (Appendix D)

Chinese-SkillSpan retains collected recruitment texts and adds model-generated annotations with documented review. The annotation prompts follow the handbook used at each stage. Encoder models offer one way to learn from these labels. Pretrained encoders and structured decoding (Devlin et al., 2019; Lafferty et al., 2001) can be combined with continued pretraining on recruitment text (Gururangan et al., 2020; Zhang et al., 2022b). SkillSpan introduced the domain-adapted JobBERT and JobSpanBERT baselines (Zhang et al., 2022b). ESCOXML-R uses taxonomy-driven multilingual pretraining (Zhang et al., 2023), and NNOSE retrieves training-data representations to support extraction, including infrequent patterns (Zhang et al., 2024). Reproducing these methods on their original tasks and adapting them to Chinese LSKT extraction require separate protocols because the languages, annotation layers, and splits differ. Other approaches use text-to-structure generation in UIE (Lu et al., 2022), distilled supervision in UniversalNER (Zhou et al., 2024), open-type encoder extraction in GLiNER (Zaratiana et al., 2024), and generative extraction with self-verification in GPT-NER (Wang et al., 2025). Their output formats and supervision differ. We therefore specify the training labels, extraction rules, and evaluation targets for each comparison, and document the data sources and intended uses following data statements and datasheets (Bender and Friedman, 2018; Gebru et al., 2021).

CORPUS AND ANNOTATION

The resource must make both linguistic decisions and annotation quality open to inspection. We therefore define the span task, distinguish data layers by their intended use, and assess independent coding separately from model-assisted review.

Task Definition and Label Schema

Given a Chinese job-ad sentence x , the task is to predict non-overlapping spans (s, e, t) . Here, s and e are zero-based, end-exclusive Unicode-code-point offsets, and $t \in \{L, K, S, T\}$. Each span must reproduce the contiguous substring $x[s : e]$, including its spelling and internal spaces. The annotation describes a requirement in the advertisement; it does not measure an applicant's ability.

Figure 1 shows how corpus preparation and annotation lead to model training and evaluation.

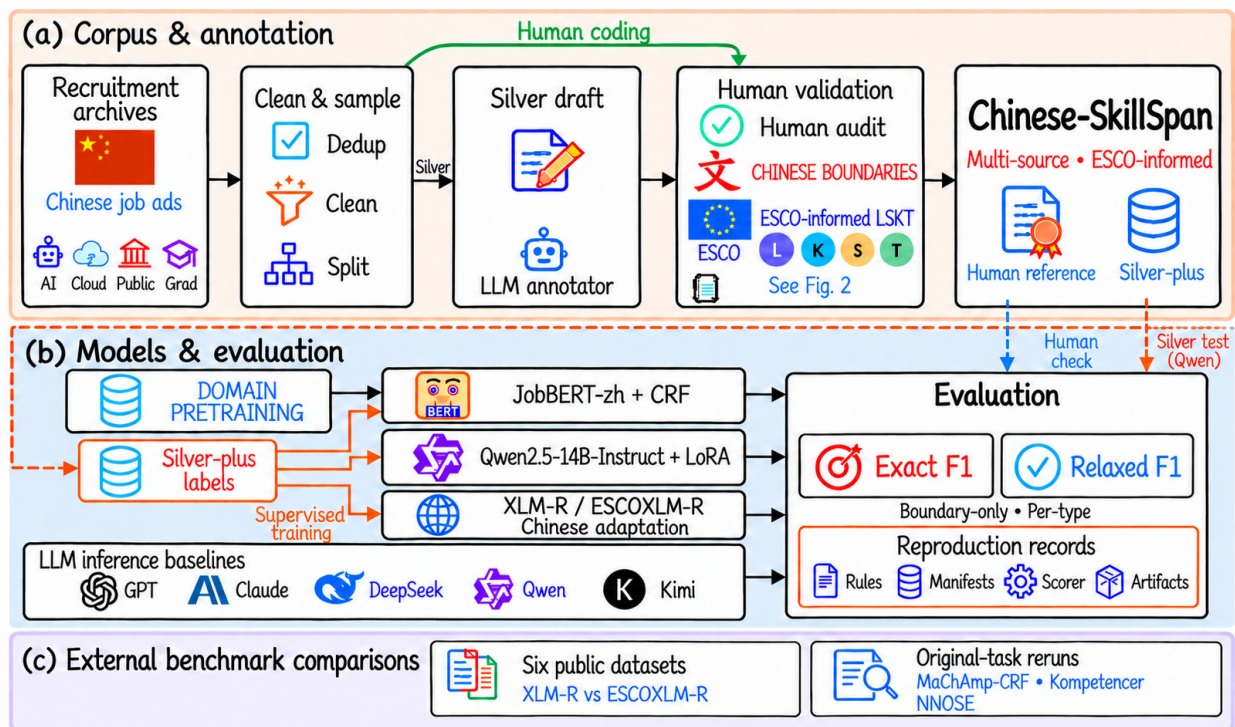


Figure 1. Chinese-SkillSpan workflow: corpus preparation, annotation, models, and evaluation. Human-reference checks and Silver-target evaluation are distinguished. External comparisons retain separate datasets, labels, and metrics. Figure 2 separates reference construction, assisted review, and the independent blinded agreement study.

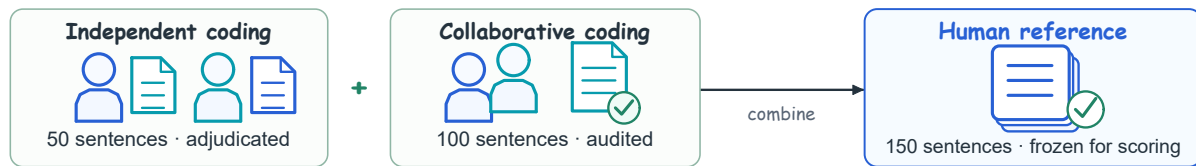
The handbook assigns L to language requirements, including language names, proficiency levels, examinations, and certificates. K covers facts, principles, theories, and fields of study. S covers occupational actions, methods, and tool use. T covers transversal capabilities such as communication, cooperation, cognition, and self-management. These categories draw on ESCO and the knowledge-skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017).

By project convention, educational and non-language qualifications also receive K. They serve as recruitment proxies for knowledge requirements, so K counts combine substantive knowledge

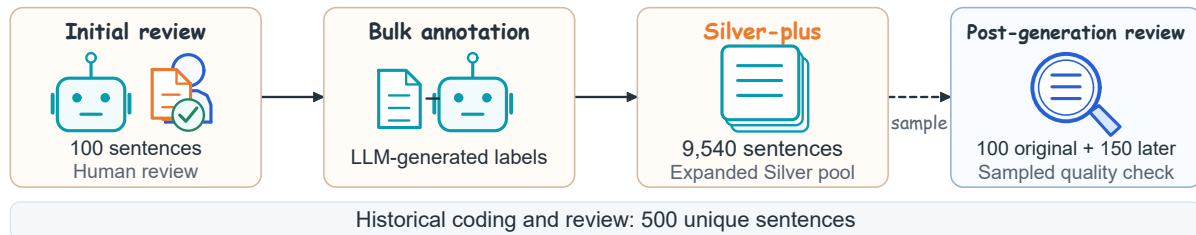
mentions with qualification proxies. We have not established separate proxy counts or evaluated the task after excluding them. The rules below explain how context determines type and how experience requirements are handled. Evaluation concerns mention boundaries and LSKT types. Human annotations provide the diagnostic reference, and Silver labels provide supervision. To test whether the handbook can be applied without shared suggestions, a separate study uses three independent coders blinded to machine and peer labels. Figure 2 separates these roles. External classification and retrieval experiments use their own datasets.

Human annotation and LLM-assisted labeling

Historical human reference



LLM annotation and sampled review



Final-handbook agreement study

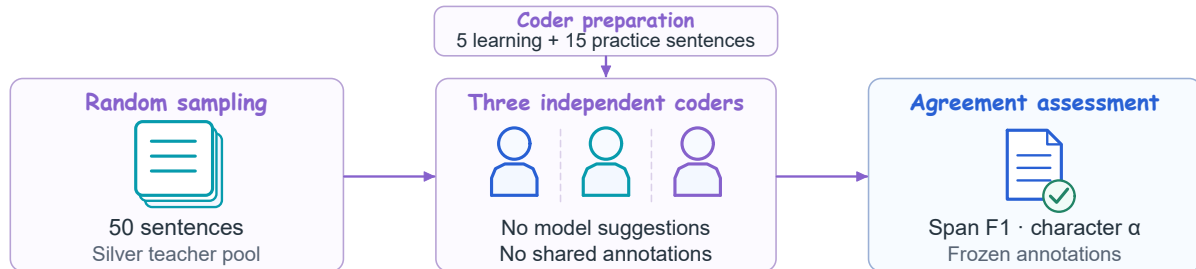


Figure 2. Distinct paths for reference construction, assisted labeling, and independent blinded agreement. The separate 50-sentence study follows training and uses three independent coders with machine and peer labels hidden. Agreement uses pre-discussion annotations. Historical coverage remains 500 unique sentences; the new sample is not added to the 150-sentence reference or used for model testing.

Corpus Construction and Annotation Layers

To cover varied recruitment settings while keeping evaluation interpretable, we distinguish the source corpus from its supervision, reference, and agreement subsets. The corpus contains 22,840 sentences from the AI, Cloud, public-sector, and graduate-recruitment collections, with an original split of 17,460 training, 2,143 development, and 3,237 test sentences. These collection names describe sources and can overlap in occupational coverage. Table 2 distinguishes the corpus from the smaller subsets used for annotation, supervision, and evaluation.

Texts came from three routes: commercial collections purchased from MacroData (MacroData, nd); public-institution recruitment announcements collected by the research team; and the Recruitment Dataset (dataset ID 163746) on Alibaba Cloud Tianchi (Tianchi, nd). For public-institution announcements, team members selected pages from government portals, university and hospital websites, and public recruitment platforms across regions of China. A crawler revisited the registered URLs weekly or monthly. Screening and cleaning produced the public-institution subset. Appendix D records the available source metadata and gaps in the original collection history. The original annotation study contains 2,601 records: a 150-sentence human reference, named Gold150 in the released files, and 2,451 records with LLM-generated Silver labels, named Silverplus. The Silver set includes 100 initially reviewed records, 730 further records from the model-disagreement queue, and 1,621 historical extraction records. The 980-record disagreement queue contributed the reference sentences and the first 830 Silver records. The human reference combines 50 calibration sentences with a non-overlapping 100-sentence challenge sample. The calibration sentences were coded independently from raw text and then adjudicated; the challenge sample was collaboratively coded and reviewed. Both informed guideline development. Their final annotations were fixed for evaluation. The reference contains 663 spans: 448 S, 125 K, 88 T, and two L. Eight sentences contain no annotated span. Its 150 complete sentence texts are distinct after NFC normalization. The two original 100-sentence review samples are separate from the reference, giving 350 unique sentences with human coding or review at this stage. The historical 15-sentence assisted-review subset is nested within this coverage. Cohort-level counts and sampling details appear in Appendices B and D.

Expanded supervision and annotation profile. We expanded supervision toward a target of 10,000 unique sentences by combining 2,064 valid unique sentences from the original files with 7,936 new candidates. The new sentences came from cloud, public-sector, and listed-company collections, with selection prioritizing requirement-related keywords. Annotation checks produced two final sets of 9,646 and 9,540 sentences. The selection and exclusion procedure is documented in Appendix D, and the corresponding experiments appear in Appendix C.

Two reviewers checked a further stratified sample of 150 newly annotated sentences, called QA150 in the released summaries. The sample contains 60 cloud, 45 public-sector, and 45 listed-company sentences and excludes the original 350 human-coded or reviewed sentences by identifier and normalized text. Both reviewers completed separate submissions with machine annotations visible. Historical human coding and review thus cover 500 unique sentences; the evaluation reference remains the original 150. The separate 50-sentence blinded agreement study is described below and is not added to this historical coverage count.

The supervision and reference subsets differ substantially in composition. Empty annotations occur in 40.4% of original Qwen training records and 48.5% of development records, compared with 5.3% of reference sentences (Table 2). Occupational skills also dominate all five profiled layers (59.7–67.6%). These differences motivate per-type and empty-sentence diagnostics alongside micro-F1. Figure 3 shows length and type distributions; Appendix B records the medians.

Annotation Guidelines and Adjudication

The guidelines address three decisions that a list of skill terms cannot settle: whether a clause is eligible, how much text to retain, and which type fits its context. Annotation begins by determin-

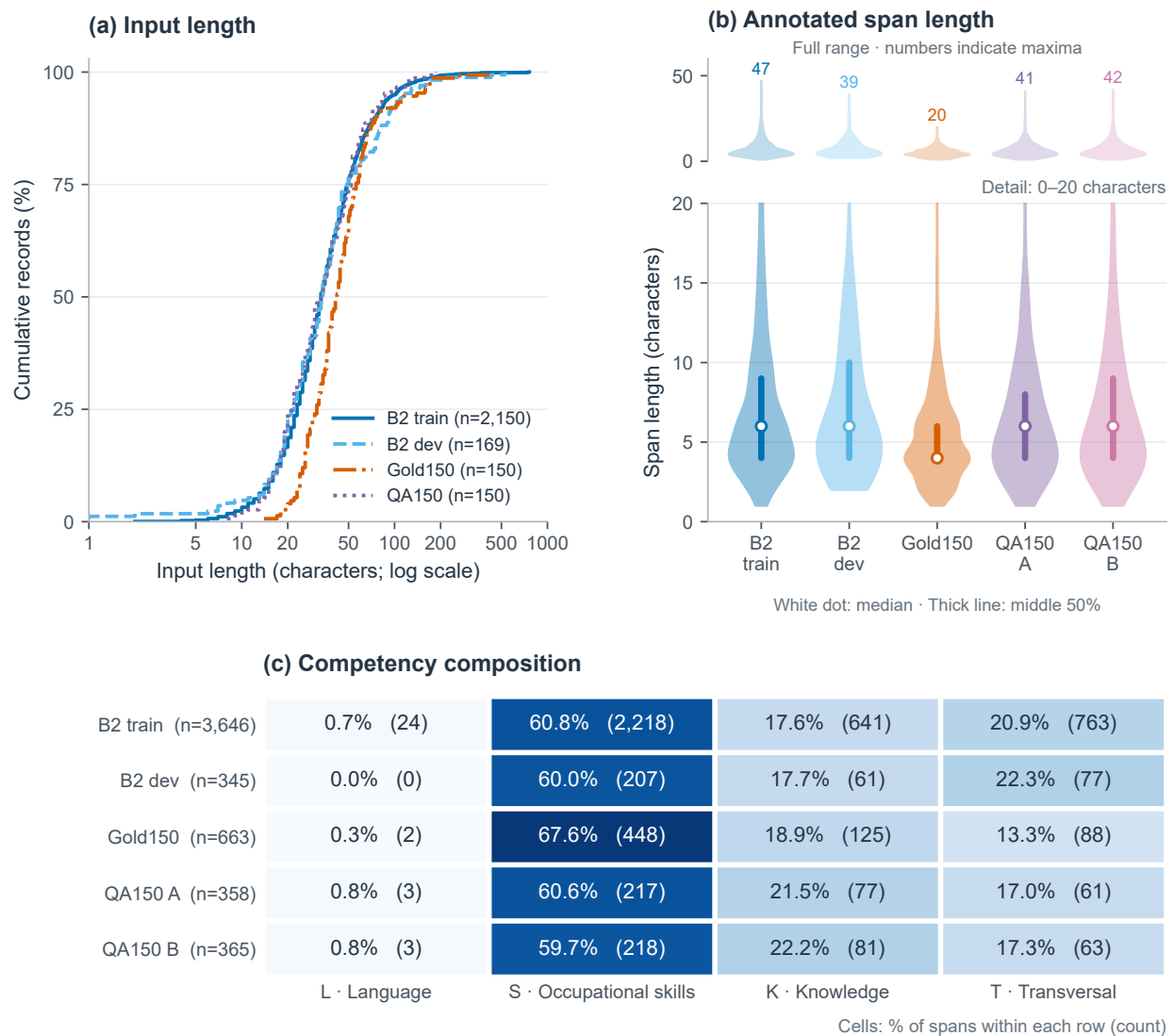


Figure 3. Benchmark annotation profile. (a) Empirical cumulative input-length distributions, including records with no spans; the horizontal axis is logarithmic. (b) Span-length violins: the main view enlarges 0–20 characters, covering 96.4–100.0% of spans in each layer; the upper overview retains the full range and labels the maxima. Both views use the same full-data density and equal maximum widths. White dots and thick lines mark medians and interquartile ranges. (c) Within-layer type percentages and span counts. B2 train/dev are the original 2,150/169-record Qwen manifests; Gold150 is the frozen 150-sentence reference. QA150 A/B are the two assisted-review layers of the same additional 150 sentences, shown once in (a). Lengths include punctuation and spaces; repeated B2 records are retained. These profiles describe historical supervision and review components; the independent three-annotator sample and the full expanded pool are not included.

Table 2. Data layers, support, and intended use. The reference and original Silver-plus pool partition the 2,601-record lineage; the expanded pool is a later experiment-specific construction. The two 100-sentence reviews are distinct subsets of Silver-plus; Appendix B distinguishes their annotation designs. Training sizes refer to actual experiment manifests; the separate agreement sample assesses coding consistency only.

Layer	Records	Support / composition	Role
Original corpus	22,840	Four source collections	Corpus-scale annotations; original split only
Human reference	150	663 spans; 8 empty; 50 + 100 cohorts	Frozen evaluation targets
Original Silver-plus pool	2,451	100 reviewed; 730 conflict-origin; 1,621 other	Candidate labels; version-specific inclusion
Initial review	100	Versioned review layers (Appendix B)	Review of LLM labels; separate from evaluation
Post-generation review	100	Machine: 144 spans; re-viewed: 148	Sampled machine–human quality check
JobBERT train / dev	2,156 / 169	Matched B1/B2 texts; different labels	Historical supervision and selection
Qwen B2 train / dev	2,150 / 169	3,646 / 345 spans; 869 / 82 empty	Earlier adapter fitting and selection
Additional dual-review QA	150	Both reviewers complete; 101 identical span sets	Quality check of the new Silver portion
Blinded agreement	50	Three independent coders; five + 15 training sentences excluded	Agreement analysis only; not a model test set
Expanded Silver (Codex)	9,540	8,842 / 348 / 350 train / validation / test	Silver-based learning and evaluation; additional Gold150 check

ing whether a clause expresses a requirement for the candidate. Benefits, applicant-side procedures, company promotion, and pure outcomes do not qualify on their own; recruitment activities performed as occupational duties are assessed separately. Mixed clauses are considered separately. A sentence with no competency mention receives an empty annotation; insufficient context or an unresolved interpretation is recorded as uncertainty.

For an eligible mention, annotators select the shortest source span that preserves the complete requirement, without a fixed length cap (Zhang et al., 2022b). Generic proficiency triggers usually remain outside the span, but a modal can remain inside a complete colloquial capability expression. Necessary action heads and complements are retained. Coordinated activities are split only when each remains interpretable; a shared action and its objects stay together. Table 3 illustrates these distinctions.

Experience expressions require a separate check. Generic work experience and duration alone are excluded. Explicit industry-background requirements use domain cores as broad-K proxies under the project convention. For occupational activities, the experience qualifier is omitted only if the remaining span preserves the requirement. It can stay outside coordinated items when each item

remains interpretable in context. Neither a conjunction nor an experience qualifier alone determines the boundary.

Type follows the local predicate and referent. Tool use receives S, while knowledge of the same technology’s principles receives K. A predicate can determine type without being part of the selected span. Familiarity alone does not decide type. General communication and coordination remain T inside duty clauses, whereas technical operations receive S. Language examinations receive L; educational and non-language qualifications receive K. Table 3 provides the bilingual cases, and Appendix A gives further examples and rule sources.

Adjudication treats scope, boundaries, and type as separate decisions (Artstein and Poesio, 2008). There is no total priority order among L, K, S, and T. Once scope and boundaries are settled, a mention with unresolved type may receive provisional S with its alternatives logged, but this is not a final training target. Same-boundary alternatives, nested candidates, and unresolved overlaps are kept in the adjudication record rather than duplicated in the flat annotation layer. Appendix A gives the full rules and their history. Each completed experiment retains the labels assigned under its original handbook.

Table 3. Chinese source examples with English explanations. Brackets mark selected substrings; subscripts give types. Glosses are display aids, not translated model inputs. These current-handbook examples do not change historical labels.

Decision	Chinese source and selected spans	English explanation
Complete boundary	熟悉 [支持服务] _s	Support services: retain the complete term. Familiarity supplies context.
Separate activities	[维护] _s 和 [支持服务] _s	Maintenance and support services: two independently meaningful activities.
Shared action	[优化曝光与转化率] _s	Optimize exposure and conversion rate: keep the action with both objects.
Contextual type	使用 [SQL] _s ; [SQL 原理] _k	Using SQL denotes skill; SQL principles denote knowledge.
Omitted referent	熟悉 [Linux 基本操作] _s 和 [原理] _k	Basic Linux operations and principles: the second span refers back to Linux without copying it (case 130).
Colloquial capability	[会聊天] _t	Being good at conversation: the modal remains within the capability expression.
Experience scope	[大项目售前] _s 经验; [机器人竞赛经验] _s	Presales activities stand without the experience qualifier; robotics-competition experience retains it.
Transversal or technical	[客户汇报] _t ; [接口对接] _s	Client reporting is communication (T); interface integration is a technical operation (S).

Human Annotation and LLM-Assisted Labeling

Human coding and model-assisted review address different quality questions. Independent coding tests whether annotators can apply the rules without a suggested answer; review checks labels proposed by a model. Historical coding informed the handbook and human reference, while the sepa-

rate three-annotator study assesses independent blinded agreement under the final handbook. For the original Silver set, an initial sample was reviewed before bulk generation. Automated checks verified substrings, offsets, labels, and coverage; human review addressed interpretation. The bulk annotations followed the handbook available at generation time. Later amendments were recorded separately, leaving completed experiments unchanged. The expanded set combined earlier supervision with additional annotations obtained through two access channels. Unresolved records were excluded. Appendices A and D identify the handbook versions, annotation channels, and inclusion decisions.

Annotation Quality and Reference Freezing

Agreement must be measured without shared suggestions if it is to assess how consistently annotators apply the handbook. We therefore report the independent blinded study in Table 4. Historical coding and assisted-review studies used different samples and rules; their scores cannot be read as a longitudinal improvement in independent reliability (Appendix B).

Blinded coding protocol. Three annotators used handbook B.sop_v4.2.14 after familiarization and training on sets of five and 15 sentences, following the training-and-feedback approach discussed by Klie et al. (2024). They then independently coded the same random sample of 50 Silver teacher-pool sentences in separate Doccano projects, blinded to both machine suggestions and one another’s labels. Training sentences are excluded from formal agreement. The sample contains one sentence from each of 50 recorded advertisement identifiers; it assesses annotation agreement and is not a new model test set. Source counts and sampling-record limits appear in Appendix B. We computed agreement from the preserved pre-discussion exports, retaining all 50 sentences and all three coder pairs. Typed exact F1 measures agreement on complete spans without designating a coder as gold (Hripcsak and Rothschild, 2005); boundary-only F1 ignores type. Nominal Krippendorff’s α uses L/K/S/T/O assignments to all 2,111 source code points. Percentile 95% intervals use 10,000 source-stratified sentence-bootstrap samples, retaining all three annotations together. These intervals condition on the observed source mixture and coders.

Table 4. Independent blinded agreement among three annotators on 50 sentences. Character coefficients are Cohen’s κ unless marked α . Brackets give 95% bootstrap intervals for typed exact F1. “Equal” counts identical sentence-level span sets. Training samples are excluded.

Comparison	<i>n</i>	Character	Typed F1	Details
<i>Final-handbook independent blinded coding</i>				
A–B	50	0.800	0.657 [0.528, 0.767]	Boundary F1 0.762; equal 26/50
A–C	50	0.848	0.739 [0.612, 0.850]	Boundary F1 0.788; equal 33/50
B–C	50	0.803	0.594 [0.459, 0.717]	Boundary F1 0.721; equal 22/50
Mean / three-rater	50	α 0.817	0.663 [0.562, 0.759]	Mean boundary F1 0.757; all three equal 21/50

Independent blinded coding leaves substantial span-level disagreement. Mean typed F1 is 0.663, below the planned 0.80 project target, which is not a universal reliability cutoff. Boundary F1 is

higher at 0.757 (95% CI [0.654, 0.846]), indicating additional type disagreement. Character-level $\alpha = 0.817$ (95% CI [0.755, 0.873]) measures a different unit and does not override these span disagreements (Artstein and Poesio, 2008). Of the 21 fully agreed sentences, ten are empty. The two L spans occur in one sentence, limiting category-level conclusions. Frozen labels and diagnostics accompany the reproduction materials.

Ethics and Data Use

The study analyzed recruitment advertisements without recruiting job applicants. Annotation and review were performed by members of the research team. Institutional ethics approval was not sought, and no formal institutional exemption is claimed. The available release documentation does not establish a field-by-field audit of employer-contact names, telephone numbers, or email addresses, and no verified removal rate is reported. The materials are provided for academic research and peer review under the reuse conditions stated in Data Availability.

EVALUATION PROTOCOL

The evaluation asks whether the resource supports informative comparisons of prompted extraction, supervised adaptation, and encoder initialization. We use a shared human reference for diagnostic comparisons and keep Silver-target and external-task results separate.

Datasets and Splits

The original supervision experiments train and select models on Silver data and score them against the fixed human reference. JobBERT uses 2,156 training and 169 development records. Six training records match three development texts after NFC normalization; removing them gives Qwen’s 2,150/169 split. Repeated texts remain within training. Neither split contains reference identifiers or complete NFC-normalized reference sentences. Partitioning is at sentence level, so different sentences from one advertisement may occur in different splits. Scoring retains the original sentence strings and character offsets.

The expanded experiments use two sets with 8,842/348/350 and 8,943/351/352 training, validation, and test sentences, respectively. Both combine earlier supervision with additional annotations, but differ in annotation channel and the examples retained after checking. Records retained after the channel replacement keep their split assignments. For Qwen, Silver-validation exact F1 selects the checkpoint, Silver-test F1 measures agreement with held-out generated labels, and the human reference provides an additional evaluation. Reference sentences are excluded from gradients and checkpoint selection. Appendix C identifies the two variants and reports their results; Appendix D documents split checks.

Evaluation Metrics

To distinguish exact extraction from approximate boundary agreement, the primary metric is typed exact-span micro-F1: a prediction matches only when both character boundaries and the LSKT type agree with the reference. Typed relaxed micro-F1 allows boundary differences through one-to-one greedy matching of same-type spans with intersection-over-union (IoU) of at least 0.5:

$$\text{IoU}(u, v) = \frac{|u \cap v|}{|u \cup v|}, \quad \text{IoU}(u, v) \geq 0.5. \quad (1)$$

Here, u and v are the sets of character positions covered by a predicted and a reference span, and $|\cdot|$ counts positions. For either matching rule, TP counts matched pairs, FP unmatched predictions, and FN unmatched reference spans, summed across sentences:

$$P_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F_{1,\text{micro}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (2)$$

All 150 human-reference sentences are scored. Expanded Qwen experiments also report scores on their Silver validation and test sets. Partially rejected responses retain accepted spans; malformed responses or responses with no accepted spans are scored as empty predictions. Empty–empty sentences add no span counts. Appendix C summarizes parser decisions and matching order and links the full scorer specification. Boundary-only scores with types collapsed are secondary diagnostics.

For experiments with three training seeds, we report the mean and sample standard deviation of the per-seed scores:

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f_i, \quad s_f = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (f_i - \bar{f})^2}, \quad n = 3. \quad (3)$$

Inference-only and expanded-pool results are single runs unless otherwise stated. The original-task reruns report their own seed counts. Scores are computed before display rounding and shown to three decimals; scientific notation preserves the sign of near-zero interval endpoints. Appendix C documents resampling procedures.

Models and Baselines

The baselines cover sequence labeling and generative extraction, so that resource behavior is not inferred from one architecture alone. JobBERT-zh combines a Chinese recruitment-domain-pretrained encoder with a CRF head. Qwen2.5-14B-Instruct (Qwen) is adapted through LoRA, using the model without a project adapter as its baseline.

The shared-guideline inference comparison includes GPT proxy, Claude Sonnet 4.5 (Claude), DeepSeek-V4-Pro (DeepSeek), Kimi k2.6 (Kimi), and Qwen. DeepSeek is evaluated with thinking enabled and disabled, giving six configurations. GPT proxy denotes the configuration accessed through a third-party service. Appendix C records service identifiers, access routes, and inference settings.

Implementation Details

Supervision and model selection. JobBERT conditions start from the same released encoder and CRF. Earlier supervision (B1 in the experiment files) uses the historical extraction labels; revised supervision (B2) combines updated guidance, LLM annotation, and adjudication. The conditions share training and development texts, with label changes in 1,440 training records and 107 development records. Development typed exact F1 selects checkpoints, so this comparison changes training supervision and selection targets together.

Qwen starts from its pretrained Instruct weights and fits a fresh LoRA adapter for each of seeds 42, 43, and 44 on the revised 2,150/169 split. Training uses a fixed system instruction, user template, and five schematic examples. Targets contain the literal span text, occurrence index, and

type. Loss applies to the assistant JSON and closing turn token. Training uses BF16, and development exact F1 selects the adapter. Table 5 gives the remaining settings. Historical JobBERT implementation gaps are documented in Appendix D.

Table 5. Recorded training settings for the original JobBERT and Qwen training comparisons. Both select checkpoints by development typed exact F1; implementation and selection records accompany the reproduction notes.

Setting	JobBERT-zh B1/B2	Qwen LoRA
Learning rate	2×10^{-5}	10^{-4}
Batch size	16	1; 16-step gradient accumulation
Maximum input length	256 tokens	8,192 tokens
Training limit	6 epochs; patience 2	6 epochs; patience 2

Qwen uses LoRA with rank 16, scaling 32, and dropout 0.05, applied to the attention q/k/v/o projections.

Inference. Each request contains one sentence’s identifier and text, together with the shared instruction and five fixed examples. A parser converts occurrence-based output into character spans. Qwen baseline and adapter inference use the same base checkpoint, prompts, parser, BF16 precision, deterministic decoding, and 2,048-token output cap. DeepSeek uses caps of 2,048 tokens without thinking and 8,192 with thinking. Full service settings and output checks are given in Appendix C.

External Benchmarks and Multilingual Encoders

To test whether domain-pretrained initialization helps consistently across recruitment tasks, the experiments compare XLM-R-large (Conneau et al., 2020) and ESCOXLM-R (Zhang et al., 2023). They use six public datasets: SkillSpan (Zhang et al., 2022b), Kompetenzer (Zhang et al., 2022a), German ICT (Gnehm et al., 2022), Green (Green et al., 2022), Sayfullina (Sayfullina et al., 2018), and FIJO (Beauchemin et al., 2022). Separate skill and knowledge streams in SkillSpan and Kompetenzer give eight extraction tasks, each retaining its original entity types.

Both encoders use fresh linear first-subword BIO heads, AdamW with a learning rate of 2×10^{-5} , batch size 8, and a maximum of 20 epochs. Training uses seeds 42, 43, and 44, with development typed exact token-span micro-F1 selecting checkpoints. For the Chinese comparison, both encoders and their new nine-label BIO heads were fine-tuned on B2’s 2,150 training and 169 development records. The executed pipeline uses Chinese source characters without translation or an English task head. It labels the first subword of each character and joins predictions from chunks of at most 128 characters, retaining the remainder of long sentences. Tokenizer placeholders preserve positions for whitespace and otherwise discarded characters; scoring uses the original text and offsets. Development F1 selects the checkpoint. Gold150 is used only for diagnostic scoring within these runs, although it was previously used in guideline development. Exact text and identifier checks found no overlap across the three partitions; advertisement-level independence is not established.

The Chinese comparison uses the same optimizer settings, maximum epochs, and three seeds as above. Its linear heads and training budget differ from JobBERT’s CRF and Qwen LoRA, so shared-reference scores support descriptive comparisons rather than a controlled ranking of those

architectures. We report typed exact and relaxed character-span F1, boundary-only F1, and macro-F1 over L/K/S/T. Macro-F1 gives L, represented by only two reference spans, one quarter of its weight.

For each external task, paired sentence-bootstrap intervals summarize the mean ESCOXML-R minus XLM-R difference conditional on the three trained seeds. We use 2,000 resamples with seed 20260921 and report training-seed SD separately. The intervals do not account for advertisement clustering or multiple comparisons.

Appendix E gives separate protocols for five-seed MaChAmp-CRF reruns on SkillSpan and German ICT, five-seed Kompetenzer classification, and single-seed NNOSE retrieval. These preserve their task-specific scoring: Danish BIO extraction differs from fine-grained classification of supplied spans, and the public German ICT release covers only part of the EDU/EXP/LNG pipeline. The datasets' languages and annotation schemes also differ, so their absolute scores do not measure relative extraction difficulty across languages.

For the Chinese encoder contrast, we additionally resampled the same 150 sentences jointly across both models and all three fixed seeds, using 10,000 draws (seed 20260924). Each draw recomputes per-seed micro-F1 and then averages the paired differences. This post hoc diagnostic interval describes reference-sentence sampling uncertainty; it does not establish population generalization or account for advertisement clustering.

RESULTS AND ANALYSIS

Inference on the Human Reference

GPT proxy achieved the highest typed exact micro-F1 among the six configurations, followed by DeepSeek with thinking disabled (Table 6). Their paired difference is 0.020, with a 95% sentence-bootstrap interval of $[-0.026, 0.064]$ (Table 7). The interval includes zero, so the point scores do not establish a stable ordering. These are comparisons of the recorded configurations, which use shared instructions but different inference budgets.

Table 6. Shared-guideline inference configurations on the human reference ($n = 150$). P/R are typed exact precision/recall. Each configuration is a single run. Inference budgets are configuration-specific. Qwen uses no project adapter.

Recorded configuration	P	R	Exact F1	Relaxed F1
GPT proxy	0.676	0.658	0.667	0.766
DeepSeek (non-thinking)	0.651	0.643	0.647	0.761
Claude	0.598	0.597	0.598	0.709
DeepSeek (thinking)	0.713	0.505	0.591	0.671
Kimi	0.581	0.563	0.572	0.714
Qwen	0.507	0.281	0.361	0.470

Both DeepSeek rows use the same model. Non-thinking and thinking denote inference modes, with output caps of 2,048 and 8,192 tokens respectively. GPT proxy denotes the proxy-accessed configuration; its recorded identifier appears in the naming guide. Cohort scores and detailed output-status counts accompany the reproduction materials.

Table 7. Paired typed exact-F1 differences on Gold150, with conditional 95% sentence-bootstrap intervals. All failures remain in the scored predictions.

Comparison	Difference	95% interval
GPT proxy minus DeepSeek non-thinking	+0.020	[−0.026, 0.064]
Qwen LoRA 42 minus no adapter	+0.138	[0.071, 0.208]
Qwen LoRA 43 minus no adapter	+0.202	[0.132, 0.276]
Qwen LoRA 44 minus no adapter	+0.197	[0.127, 0.271]
Qwen LoRA seed mean minus no adapter	+0.179	[0.113, 0.249]

Fine-Tuning with Silver-plus Labels

All three Qwen adapters improved exact precision and recall over the no-adapter baseline under the same inference protocol (Table 8, panel A). Mean typed exact micro-F1 increased from 0.361 to 0.540 ± 0.035 . Each seed’s paired interval against the baseline also excludes zero (Table 7), conditional on the recorded runs and reference sentences. This consistency across seeds supports end-to-end task adaptation with the supplied Silver supervision on the selected reference. It does not isolate semantic extraction from output-format learning: partially rejected responses decrease from 69 without an adapter to 2, 4, and 1 in the three adapted runs, under the same parser.

Table 8. Fine-tuning comparisons on the human reference ($n = 150$). Typed exact micro-F1 is reported as mean $\pm$ sample SD over training seeds 42, 43, and 44; the Qwen no-adapter baseline is a single run.

A. Qwen: fixed inference protocol

Condition	Train / dev	Exact F1
No project adapter	—	0.361
B2 LoRA	2,150 / 169	0.540 ± 0.035

B. JobBERT-zh: historical supervision and selection

Supervision and selection labels	Train / dev	Exact F1
Earlier (B1)	2,156 / 169	0.142 ± 0.014
Revised (B2)	2,156 / 169	0.554 ± 0.005

A: same base checkpoint, frozen messages, occurrence schema, parser, and decoding settings; B2 development F1 selects each adapter. B: same initial encoder/CRF and matched training/development texts, but both sets of labels change. The split and checkpoint limitations are summarized in Limitations and Responsible Use. Per-seed Qwen results are retained in the run ledger.

The historical JobBERT configuration achieved higher reference agreement with revised supervision (Table 8, panel B). The two conditions start from the same encoder and CRF and use matched training and development texts, but revise both sets of labels. Their difference therefore reflects changes to supervision and checkpoint selection together.

Supplementary Analyses

The Qwen improvements extend across the three reported span-length bins: every adapter increases exact recall in each bin (Figure 4). False-positive spans on the eight empty-reference sen-

tences also decrease, although the changes vary by seed. Type-level F1, span support, and separate sentence and span counts are shown in the figure.

Exploratory experiments with expanded supervision appear in Appendix C. Those runs change the source composition, annotation targets, and retained examples together. They do not isolate an expansion effect. The lower expanded JobBERT scores remain unexplained by the available diagnostics.

Chinese Encoder Baselines and External Comparisons

The Chinese comparison does not establish a reliable advantage of ESCO initialization. The encoders achieved mean exact F1 of 0.522 (XLM-R-large) and 0.538 (ESCOXLM-R), with all six prediction files independently rescored against the original reference (Table 9). The mean difference was 0.016; its paired 95% interval was $[-0.003, 0.034]$. This interval includes zero. Higher relaxed and boundary-only scores identify sensitivity to boundary and type matching, without attributing errors to a particular construction. Four-type macro-F1 is 0.607 and 0.622; excluding the two-span L class gives descriptive K/S/T macro values of 0.542 and 0.562, respectively. This sensitivity does not replace the stated four-type metric.

The archived run summaries show mixed effects of domain-pretrained initialization across the external tasks (Table 10). ESCOXLM-R had a higher mean typed exact F1 on SkillSpan skills and Sayfullina and a lower mean on the other six streams. The largest decreases were on SkillSpan knowledge and German ICT; only these two pointwise bootstrap intervals excluded zero (Appendix E, Table 19). All seeds were retained, including runs selected early in training. FIJO's comparison is based on only 50 test sentences.

The original-task reruns answer related but separate questions. Combining English and Danish training improves Danish classification macro-F1 for both tested encoders. NNOSE retrieval improves the single-seed Green results but not SkillSpan or Sayfullina. Appendix E reports these experiments with the public-subset MaChAmp reruns, separating classification, retrieval, and extraction scores according to their protocols.

The datasets, scorers, and training architectures differ, so these absolute scores do not establish a cross-language difficulty ranking or isolate initialization effects across the task-specific protocols.

Table 9. Supervised adaptation of multilingual encoders on Gold150 (0–1), mean $\pm$ sample SD across three seeds.

Encoder	Typed exact	Typed relaxed	Boundary	Macro-F1
XLM-R-large	0.522 ± 0.022	0.650 ± 0.021	0.571 ± 0.022	0.607 ± 0.016
ESCOXLM-R	0.538 ± 0.021	0.662 ± 0.022	0.582 ± 0.024	0.622 ± 0.010

B2 training/development: 2,150/169 records. Fresh linear LSKT head; official `cnss-lsht-1.2.0` scorer; 150 human-reference records. Boundary is exact span matching with types collapsed; macro-F1 averages L/K/S/T. Gold150 is not a new blind test and contains only two L spans. All three seeds (42/43/44) are included; per-seed scores and selected epochs are retained in the model audit archive.

Table 10. Common-protocol typed exact token-span micro-F1 (0–1), mean $\pm$ sample SD across three seeds.

Dataset	Stream	Train/dev/test	XLM-R-large	ESCOXLM-R
SkillSpan	Skill	4,800/3,174/3,569	0.567 ± 0.012	0.577 ± 0.006
SkillSpan	Knowledge	4,800/3,174/3,569	0.698 ± 0.021	0.672 ± 0.052
Kompetencer	Skill	778/346/262	0.545 ± 0.028	0.527 ± 0.004
Kompetencer	Knowledge	778/346/262	0.618 ± 0.035	0.593 ± 0.044
Gnehm ICT	ICT	19,889/2,332/2,557	0.891 ± 0.005	0.867 ± 0.032
Green	Native types	8,669/964/335	0.493 ± 0.009	0.492 ± 0.007
Sayfullina	Skill	3,705/1,855/1,851	0.920 ± 0.004	0.925 ± 0.005
FIJO	Native types	399/50/50	0.343 ± 0.019	0.339 ± 0.007

Both encoders use fresh linear BIO heads, 20 epochs, and development-set checkpoint selection. SkillSpan and Kompetencer have separate skill and knowledge streams; Green and FIJO retain their entity types. German data use the public ICT release. The shared protocol differs from the original CRF and NNOSE methods.

Boundary and Type Disagreements in Human Coding

The three-annotator study shows that character agreement alone understates the difficulty of reproducing complete typed spans (Table 4). Boundary-only agreement exceeds typed agreement, so disagreements concern both extent and type. Historical calibration cases explain why the handbook distinguishes job-use tools from knowledge and general communication from technical operations; they do not establish the distribution of errors in the separate blinded sample. These findings support retaining both span-level and character-level agreement measures.

Implications for Benchmark Use

Exact–relaxed F1 gaps make boundary placement a substantive benchmark outcome. A plausible competency phrase may still have the wrong extent. These aggregate scores motivate boundary and type diagnostics but do not identify the most error-prone constructions; that requires adjudicated mismatch analysis.

Agreement with generated labels does not by itself establish agreement with human annotations. The expanded Qwen run achieves 0.809 on its Silver test and 0.585 on the human reference (Table 16). The two sets differ in both texts and targets, so the gap cannot estimate Silver-label error, but it illustrates why both evaluations are needed.

Matched Qwen runs provide the clearest adaptation evidence because inference is fixed. JobBERT and expansion studies jointly change labels, selection, or source composition, so they cannot isolate those effects. Type, length, and empty-sentence diagnostics qualify the aggregate comparisons.

LIMITATIONS AND RESPONSIBLE USE

Reference design and coverage. The calibration/challenge reference informed guideline development; exclusion from gradients and checkpoint selection does not remove that dependence. Its small, uneven type distribution and keyword-prioritized expansion limit generalization. Independent blinded coding yields mean typed F1 of 0.663, with only two L spans. The 50-sentence

teacher-pool sample does not establish corpus-wide reliability, and historical assisted review is a different design. Annotations describe advertised requirements, not applicants' abilities.

Data access and reproducibility. Public labels and predictions support inspection and rescoring, but the complete expanded-pool training inputs remain unavailable. Collection records, historical implementation details, and label-version compatibility checks are incomplete, limiting retraining and verification. Component-level access and reuse terms are specified below. Further applications require source permissions, privacy safeguards, human oversight, and external validation.

Experimental comparisons. Sentence-level splits, residual JobBERT train/development matches, and incomplete CRF selection history limit independence claims. Expanded-pool studies use single seeds; inference budgets differ, and some proxy model identities remain unverified. External tasks differ in labels, splits, scorers, and selection. Bootstrap intervals condition on trained seeds without advertisement clustering or multiple-comparison adjustment. These results characterize tested settings, not a general ranking of models or languages.

CONCLUSION

Chinese-SkillSpan specifies how to delimit and type competency mentions in Chinese recruitment text, with explicit treatment of coordination, contextual types, and negative cases. Its contribution is a versioned resource and an inspectable annotation procedure. Independent coding by three annotators, blinded to machine and peer labels, measures final-handbook consistency separately from historical coding and assisted review. Its remaining boundary and type disagreements show where adjudication and rule clarification are still needed.

On the development-used human reference, Qwen adaptation improved end-to-end extraction under fixed inference rules. The additional Chinese-supervised encoders provide reproducible alternatives, but the paired interval does not establish an ESCO initialization advantage. Boundary-sensitive scores and the type, length, and empty-sentence diagnostics identify where further assessment is needed. These findings support diagnostic use of the released resource. The blinded agreement results provide evidence about the annotation process, but do not establish model generalization to new advertisement documents or resolve all historical label-version differences.

DATA AVAILABILITY

The dataset archive (v0.1.3; DOI: 10.5281/zenodo.22698504) contains the corpus, handbook, 150-sentence human reference, and Silver-plus B2 training/development files; 10.5281/zenodo.22288337 is the concept DOI. The historical annotation-layer release separately provides pseudonymized coding and QA150 exports with original text and offsets. The independent blinded three-annotator study provides frozen labels, source identifiers, agreement code, and results. These later annotation studies are outside the original Zenodo archive.

The Qwen reproduction archive (DOI: 10.5281/zenodo.22851581) contains three B2 LoRA adapters, selected expanded-Silver checkpoints, predictions, configurations, and scoring tools. The complete texts and labels for the final 9,540- and 9,646-record pools remain outside the public archives, so the expanded experiments cannot be fully retrained from public inputs alone. The six Chinese-supervised XLM-R/ESCOXLM-R checkpoints are linked in the Hugging Face model index; their code, configurations, predictions, and verification records are in the audit archive (DOI: 10.5281/zenodo.22937245). The evidence index distinguishes local rescoring from server-reported checkpoint inference.

The JobBERT-zh initialization supplies the domain-adapted encoder and inherited CRF; the B2 continuation supplies continued checkpoints, with seed 42 as its default. The project repository and reproduction guide link code, preprocessing, scoring, and experiment records. Table 17 summarizes access and verification; Appendix E links external-benchmark protocols and results. Reuse terms vary by component. Explicitly covered software and documentation retain Apache-2.0. Author-controlled annotations and research materials permit academic research and peer review with attribution; this permission excludes third-party recruitment text. Model checkpoints and adapters follow their accompanying notices and base-model licences. The data-access guide specifies these terms; Appendix D records earlier versions and access limits.

APPENDIX A. CODING GUIDE AND EXAMPLES

This guide follows Handbook v4.2.14 and complements Table 3. The October 2 Chinese reader edition aligns explicit examples with the manuscript; cases without a determinate answer remain unresolved. It is a documentation revision, not a replacement for frozen study inputs. The archived bilingual manuals and crosswalk accompany the source package; Section A.4 records version boundaries.

A.1 Label definitions and context tests

Square brackets mark selected source substrings, with a separate type code after each span. Brackets, colours, and type codes are display aids and are not source characters; boundaries remain identifiable in grayscale. Examples illustrate the handbook, not an additional evaluation sample. *Source key.* E: ESCO's skills hierarchy and transversal concepts (European Commission, 2024); Q: EQF definitions (Council of the European Union, 2017); Z: related annotation guidance in SkillSpan, Appendix B (Zhang et al., 2022b); I: unitization and categorization in agreement analysis (Artstein and Poesio, 2008); P: our V4.2.14 operational decision. Combined marks indicate adaptation, not a verbatim source rule. Chinese examples and their exact boundaries are project illustrations.

Table 11. Current LSKT decisions: definitions, examples, and contextual contrasts.

Label	Operational definition	Chinese example
L: Language	Natural-language proficiency, language names, levels, examinations, and certificates. ^[E,P]	[英语六级] [L]; [日语 N2] [L]
K: Knowledge	Facts, principles, theory, and fields of knowledge. ^[Q] Education and non-language qualifications map to K by project convention. ^[P]	[计算机专业] [K]; [本科及以上学历] [K]
S: Occupational skill	Application of a tool or method, or an occupational activity indicated by the context. ^[Q,P]	使用 [SQL] [S]; [数据分析] [S]
T: Transversal competence	General communication, cooperation, cognition, self-management, and dispositions. Duties may also express T. ^[E,P]	[客户汇报] [T]; [责任心] [T]
Context test	Use versus principles distinguishes S from K; general coordination differs from a technical operation. No fixed label priority applies. ^[Q,E,P]	了解 [SQL 原理] [K]; [协调外部资源] [T]; [接口对接] [S]

A.2 General annotation decisions

G1–G15 are quick-reference identifiers. Their source sections and amendment IDs (R08–R23) are mapped in the release; they do not replace the handbook’s rule numbering.

Table 12. Annotation principles for target identity and complete source boundaries.

Rule	Annotation principle	Chinese example
G1	The annotation scheme distinguishes L, K, S, and T. Collapsed categories are used only in secondary evaluation. ^[E,P]	[英语六级] [L]; [计算机专业] [K]; [数据分析] [S]; [责任心] [T]
G2	Final annotations contain non-overlapping spans. A sentence may contain several independent mentions, with each character assigned to at most one span. ^[P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 No additional covering list span.
G3	Each mention is a contiguous source substring that preserves spelling and internal spaces. Any normalization is stored separately. ^[Z,P]	熟悉 [Jekins] [S] Keep the source spelling, not Jenkins.
G4	Annotators select the shortest span that expresses a complete requirement. Words and identifiers remain intact; completeness takes priority over length, with no fixed length cap. ^[Z,P]	[支持服务] [S] Not the fragment 支持服.
G5	Generic proficiency triggers usually remain outside the span, while heads needed to express the requirement remain inside. Excluded triggers can still inform type assignment. ^[Z,P]	会使用 [Excel] [S]; [会聊天] [T]; 有 [服务意识] [T]

Table 12 (continued)

Rule	Annotation principle	Chinese example
G6	Coordinated items are separated when each remains meaningful in context. Shared qualifiers may stay outside the spans, and implied words are not added. [Z,P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 Shared-experience case: see A.3.
G7	A necessary action, object, or shared head remains within the span when splitting would change its meaning. Span length alone does not determine type. [Z,P]	[优化曝光与转化率] [S] Isolated metric names would lose the optimization activity.

Table 13. Annotation principles for scope, experience, type, and unresolved records.

Rule	Annotation principle	Chinese example
G8	Benefits, company promotion, administrative instructions, experience durations, and pure outcomes are outside competency scope on their own. Mixed clauses are assessed separately. [Z,P]	五险一金, 带薪年假 → []; [分析推广效果] [S], 保证流量/用户增长
G9	Experience wording is omitted only when the remaining activity preserves the requirement. Necessary practical-experience wording is retained. Explicit industry background follows the handbook's broad-K convention. Experience wording is never added to the source. [P]	[机器人竞赛经验] [S] Retain the suffix if a bare competition name changes the requirement; do not label generic 经验 alone.
G10	Education qualifications include their threshold wording. Language examinations and certificates receive L; non-language qualifications receive K. [Z,P]	[本科及以上学历] [K]; [大学英语 6 级] [L]; 了解 [ISO 27001] [K] Standard familiarity, not a claim of personal certification.
G11	Tool and method mentions are typed from their predicate context. Occupational use supports S; principles or theory support a complete K phrase. Familiarity alone does not determine type. [Q,P]	使用 [SQL] [S]; 了解 [SQL 原理] [K]
G12	Occupational functions are distinguished from general cognition, communication, and coordination. A duty clause may contain T; usefulness across industries alone does not establish T. [E,P]	[编码能力] [S]; [沟通能力] [T]; [协调外部资源] [T]
G13	Scope, boundaries, and type are resolved separately, without a total label priority. Once scope and boundaries are valid, type-only uncertainty may receive provisional S with alternatives logged; this is excluded from final training targets. [I,P]	Heading alone: 能力提升 Insufficient scope evidence: record as unresolved, not a confirmed empty negative.

Table 13 (continued)

Rule	Annotation principle	Chinese example
G14	Offsets count original Unicode code points, beginning at zero and excluding the end position. The indexed substring must reproduce the mention exactly; segmentation is a derived view. It does not replace the fixed human-reference offsets. ^[P]	text[start:end] == span 句子原文、内部空格与偏移保持一致。
G15	Alternatives and overlaps are recorded separately from the final flat layer. An unresolved record is excluded in full from training, with confirmed empty, missing, and unresolved annotations distinguished. ^[P]	五险一金 → confirmed []; 能力提升 without context → unresolved. Known partial spans do not make an unresolved record training-ready.

A.3 Shared experience: split only when meaning survives

The V4.2.14 amendment (R23) clarifies a shared-experience construction. ^[P] Each occupational item below remains identifiable in context; the shared experience qualifier stays outside. Do not retain a covering span as well.

4、三年以上大中型 [互联网公司经营] [S] 或 [BI] [S] 相关工作经验, [互联网公司广告运营分析] [S] 优先;

Three or more years of operations or BI experience in large or medium-sized internet companies; internet-company advertising operations analysis preferred.

Selected offsets: [9, 16), [17, 19), and [26, 37). BI is S in this context, not by a universal dictionary rule. In contrast, G7 retains the shared action in 优化曝光与转化率; splitting must not leave bare outcomes. The earlier long span in the initial reviewed LLM-annotation dataset remains historical rather than being silently overwritten.

A.4 Version boundaries

The human-reference metadata identify v4.2.10, while the original bulk-generation prompt follows v4.2.12. The current handbook is v4.2.14. Later amendments do not retrospectively change completed annotations. Inspection confirmed v4.2.10 metadata on all 150 reference records; the B2 rows do not carry individual handbook-version fields. One documented difference is present in B2 training record 1838-s0008: its historical two-span annotation retains a shared experience phrase that R23 later splits into three spans. This named record is absent from Gold150 and B2 development. A rule-family screen covers all 2,469 reference, training, and development records, but flags candidates for contextual review rather than proving semantic compatibility. The original targets and scores remain unchanged. Table 3 and Tables 11–13 illustrate current decisions; they are teaching materials rather than a relabeling of historical data.

The version ledger records experiment-specific protocols, identifiers, hashes, and unresolved bindings. The accompanying manuals preserve original and revised spans and their rationale. Human-reviewed Silver-plus remains supervision unless separately designated and isolated for evaluation. Evaluation uses the frozen reference, shared protocol, occurrence schema, parser, and scorer; the reproduction guide maps these terms to their released files (Appendix D).

APPENDIX B. ANNOTATION-QUALITY EVIDENCE

Historical coding and assisted-review comparisons below use designs distinct from the independent blinded study in Table 4. The initial 50-sentence calibration study involved independent raw-text coding before adjudication under the historical handbook. The sentences were sampled from the 980-record conflict queue. Available exports reproduce 98 exact matches between 211 and 230 spans, giving F1 0.444. Character-level κ is 0.567 for L/K/S/T plus O, over all 2,571 source code points, including spaces and punctuation. All 50 identifiers belong to the calibration portion of Gold150 and none occurs in the B2 training or development identifiers. Thus, the study supplies independent-coding evidence, but is not an additional test cohort. Reproduction from later exports does not establish byte identity with the original frozen files. The calculation record preserves the arithmetic and supporting counts.

The reference's challenge cohort contributes 415 spans (290 S, 83 K, 41 T, one L), and the calibration cohort contributes 248 (158 S, 42 K, 47 T, one L). Seven challenge sentences and one calibration sentence have no annotated span. The identifiers contain 61 document prefixes; these are identifier-based groups rather than independently reconstructed advertisements. The adjudicated reference remains separate from the original coding and assisted-review layers.

Character-level Cohen's κ adjusts observed agreement for agreement expected from the two coders' marginal label frequencies:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \sum_{c \in \mathcal{C}} p_A(c) p_B(c). \quad (4)$$

Here, p_o is the fraction of source character positions with identical labels, and $p_A(c)$ and $p_B(c)$ are the two coders' label proportions. The label set $\mathcal{C}$ comprises L/K/S/T and O. For the four-label scheme plus O, the available exports yield $p_o = 0.771$ and $p_e = 0.471$, giving $\kappa = 0.567$. These character-level quantities are distinct from the span counts in Eq. (2); the 2,571 positions are nested within 50 sentences, not 2,571 independent samples.

Empty-empty sentences contribute no span true positives. Span matching and character-level agreement use different units; optional category projections are documented in the reproduction materials.

The post-generation review sample instead compares a machine-prefilled layer with human review: 140 exact typed matches among 144 machine and 148 reviewed spans give P/R/F1 of 0.972/0.946/0.959. Ninety-five sentences have identical complete span sets. The sample was stratified by historical source, empty/nonempty status, and text length from 2,179 eligible records; two small strata totalling 11 records were unsampled. The estimate describes this reviewed sample, not an unbiased corpus-wide accuracy estimate.

The 15-sentence auditor subset is nested within the post-generation Silver review cohort. Both annotators had access to machine coding suggestions. Their human layers contain the same 27 spans, yielding exact span F1 and character kappa of 1.000. These values describe agreement under machine-assisted review, where shared suggestions can influence both annotators; they are not evidence of independent blind coding or corpus-wide reliability under the final handbook.

Historical annotation profile. Median input lengths are 33 Unicode code points in each B2 supervision split, 41.5 in the reference, and 33.5 in the additional review sample. Median span length is six code points in supervision and both review layers, compared with four in the reference. Counts include spaces and punctuation; Figure 3 shows these historical distributions.

Dual-Reviewer QA on the Expanded Silver Sample

The completed export contains both human layers for the 150 preselected sentences, with all confirmations recorded separately for each reviewer. Identifiers exactly match the stratified draw, and normalized sentence texts have no overlap with the original 350. The sample includes 60 cloud, 45 public-sector, and 45 listed-company sentences. Machine suggestions were visible, so separate submissions provide reviewer–reviewer agreement under assisted review rather than blind agreement.

Table 14. Paired agreement in the additional QA150 sample. Exact typed span F1 uses original pre-adjudication span sets. Intervals are percentile 95% confidence intervals from 10,000 sentence resamples within source strata (seed 20260917). They are conditional on the sampled source quotas.

Comparison	Exact F1	95% interval	Identical sentences
Reviewer A–Reviewer B	0.816	[0.769, 0.860]	101/150
Astra–Grok suggestions	0.962	[0.937, 0.983]	138/150
Reviewer A–Astra	0.823	[0.776, 0.866]	100/150
Reviewer B–Astra	0.850	[0.803, 0.892]	108/150

The 49 sentences with differing human span sets remain part of the raw agreement calculation. The exported annotations are not replaced by their intersection or a consensus layer. Character-level calculations assign O/L/K/S/T to original Unicode-code-point positions, including spaces and punctuation. Two positions have conflicting types within one reviewer layer and are excluded from both character sequences, leaving $N = 5,990$ positions. All original spans remain in span-level F1. These raw review layers are not substituted for Gold150; the affected record is identified in the accompanying calculation record. The available records do not establish a final adjudication or record-level propagation into each later training target. This unresolved provenance is separate from the documented raw-layer agreement calculation. The resulting observed agreement is $p_o = 0.934$, with $p_e = 0.439$ and $\kappa = 0.883$. For nominal Krippendorff’s α , let m_c count assignments to category c across both reviewers, with $\sum_c m_c = 2N$. The observed and expected disagreements are

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = 1 - p_o, \quad D_e = 1 - \frac{\sum_c m_c(m_c - 1)}{2N(2N - 1)}. \quad (5)$$

Here $D_o = 0.066$ and $D_e = 0.560$, giving $\alpha = 0.883$. The character coefficients and span F1 describe different units of agreement (Artstein and Poesio, 2008); the many O positions contribute to the character calculation. Agreement between two machine suggestion layers likewise measures concordance, not accuracy against an independent human reference. Export hashes, calculation code, and detailed diagnostics accompany the reproduction record.

Independent blinded coding by three annotators. The formal sample contains eight listed-company, 16 public-institution, and 26 Tianchi sentences. Random sampling and the sequence of familiarization, training, and independent coding were confirmed by the authors. The archived exports establish sample membership and source counts, but the complete candidate frame and sampling execution log were not included in this release. The original draw therefore cannot be independently reproduced from the release; source-stratified bootstrap resampling is an analysis

step, not evidence of the draw procedure. Five- and 15-sentence training sets are excluded from formal agreement. The analysis uses pre-discussion labels, not later edited exports. The October 2 reader-edition wording changes did not trigger a new coding round. The three coders in Table 4 produced 97/113/106 spans; exact typed matches numbered 69/75/65 for A–B/A–C/B–C. Pairwise F1 ranges are 0.603–0.757 for K, 0.475–0.659 for S, and 0.722–0.837 for T. All coders identified the same two L spans in one sentence. O accounts for 59.5–64.5% of character positions. The linked reproduction materials retain the training records, freeze receipts, later-edit comparisons, per-type counts, and complete bootstrap specification; none of the 10,000 resamples produced an undefined reported coefficient.

APPENDIX C. SUPPLEMENTARY EVALUATION

This appendix documents the shared evaluation protocol, uncertainty and error diagnostics, and exploratory expanded-pool results. The supporting records provide per-run ledgers, detailed scorer behaviour, and historical protocol comparisons. Those histories use different output formats and demonstration settings and are not treated as controlled estimates of prompt effects.

Shared-Guideline Evaluation against the Human Reference

The shared protocol evaluates six configurations on all 150 reference sentences, using V4.2.12-based general instructions and five schematic examples.

Configuration and output records. Table 15 distinguishes the original annotation teacher, inference baselines, and expansion channels. It lists recorded service identifiers and access routes without independently establishing provider checkpoints.

Models return an ID and spans as literal text, zero-based occurrence, and L/K/S/T type. The span parser resolves spans to Unicode offsets for exact and relaxed scoring with the released span scorer. All supplied input texts match the frozen reference.

Qwen uses deterministic decoding; Kimi uses non-thinking mode at temperature 0.6. DeepSeek output caps are 2,048 tokens without thinking and 8,192 with thinking. Access routes appear in Table 15; full configurations accompany the reproduction notes.

Output validity. Qwen without an adapter has 69 partially rejected responses, compared with 2, 4, and 1 for the three adapters. DeepSeek thinking has 18 whole-response format failures. These counts do not separate formatting effects from other contributions to F1. All outcomes remain included in scoring under the parser rules; the complete output-status ledger retains every configuration and failure category.

Parser and matching conventions. The parser resolves literal spans to source offsets, sorts candidates by start offset and original output order, and keeps the first compatible candidate. Later duplicates, same-boundary type conflicts, and overlaps are rejected. Partial responses retain accepted spans; fully rejected or malformed responses become empty predictions. Identifier coverage is checked before scoring.

Exact matching deduplicates identical span tuples within each sentence. Relaxed matching traverses reference spans in source order and selects the unused same-type prediction with highest IoU, taking the first prediction on a tie. The threshold is 0.5, and matching is greedy rather than globally optimized. Empty–empty records contribute no span counts. The scorer specification

Table 15. Model identity and access roles. Recorded service identifiers do not independently establish a provider’s underlying checkpoint.

Role	Recorded identifier	Access and configuration
Original teacher	<code>gpt-6-astra</code>	Recorded annotation identifier; distinct from the evaluated GPT proxy. Exact provider checkpoint unverified.
GPT baseline	<code>gpt-5.6-terra</code>	Third-party proxy; identity unverified.
Claude baseline	Claude Sonnet 4.5	Proxy access; provider checkpoint unverified.
DeepSeek baseline	<code>deepseek-v4-pro</code>	Provider endpoint; non-thinking/thinking, 2,048/8,192-token output caps.
Kimi baseline	Kimi k2.6	Provider endpoint; non-thinking; temperature 0.6.
Qwen baseline / LoRA	Qwen2.5-14B-Instruct	Local weights; deterministic decoding; output cap 2,048. Original Hub commit not recorded; archived file hashes identify the supplied artifacts.
JobBERT-zh	Released initialization and B2 continuation	Local encoder and inherited CRF; initialization-selection history incomplete.
Expansion annotation	Codex CLI / API-compatible proxy	Separate access channels; neither denotes a single independently verified model version.

The shared protocol was frozen for September 10 execution; the GPT completion record includes a September 11 retry. Per-call metadata, where retained, accompany the raw records. Unspecified dates, decoding fields, and checkpoint identities are not inferred from current service names.

preserves schema validation, BIO edge cases, and executable checks. The sentence-level counts reproduce the published exact scores before resampling.

Conditional resampling. We jointly resampled the 150 reference sentences across configurations 10,000 times with seed 20260923. Typed exact micro-F1 was recomputed from sentence-level span counts. For the LoRA mean, the same resampled sentences were used for all three fixed training seeds before averaging F1. Table 7 reports percentile intervals for paired differences. These post hoc intervals condition on the development-used reference and recorded runs. They do not model advertisement-level clustering or adjust for multiple comparisons. Seed SD describes a separate source of variation.

Observed mismatch examples. In the frozen no-adaptor Qwen output for 1801-s0004, the predicted S span [5, 24) is 参与公司无人机飞控系统技术项目开发策划, whereas an overlapping reference S span is [16, 24), 技术项目开发策划. For 1802-s0005, both layers select 方法 at [6, 8), but the prediction is S and the reference is K. These are actual prediction–reference differences, not handbook teaching examples; they do not establish that either local wording alone resolves the contextual annotation decision. After removing exact matches, the recorded pairing procedure identifies 27 same-boundary/different-type pairs, 89 same-type/overlapping-boundary pairs, and 65 other unmatched predictions, totalling 181 false positives. These structural categories do not establish linguistic causes or adjudicated scope errors. The mismatch ledger records the pairing order and parser errors; a full adjudicated error taxonomy remains unavailable. Offline rescoring reproduces the exact and relaxed results for all six inference configurations and

all three Qwen LoRA runs. The reproduction notes distinguish the available scoring artifacts from the requirements for rerunning model inference.

DeepSeek in thinking mode shows higher precision and lower recall than the non-thinking configuration under their respective output budgets.

Per-seed Qwen results. The per-seed record retains precision, recall, exact/relaxed F1, and calibration/challenge scores underlying Tables 8 and 7. Figure 4 diagnoses the same frozen predictions.

Matched Qwen Frozen-Prediction Diagnostics

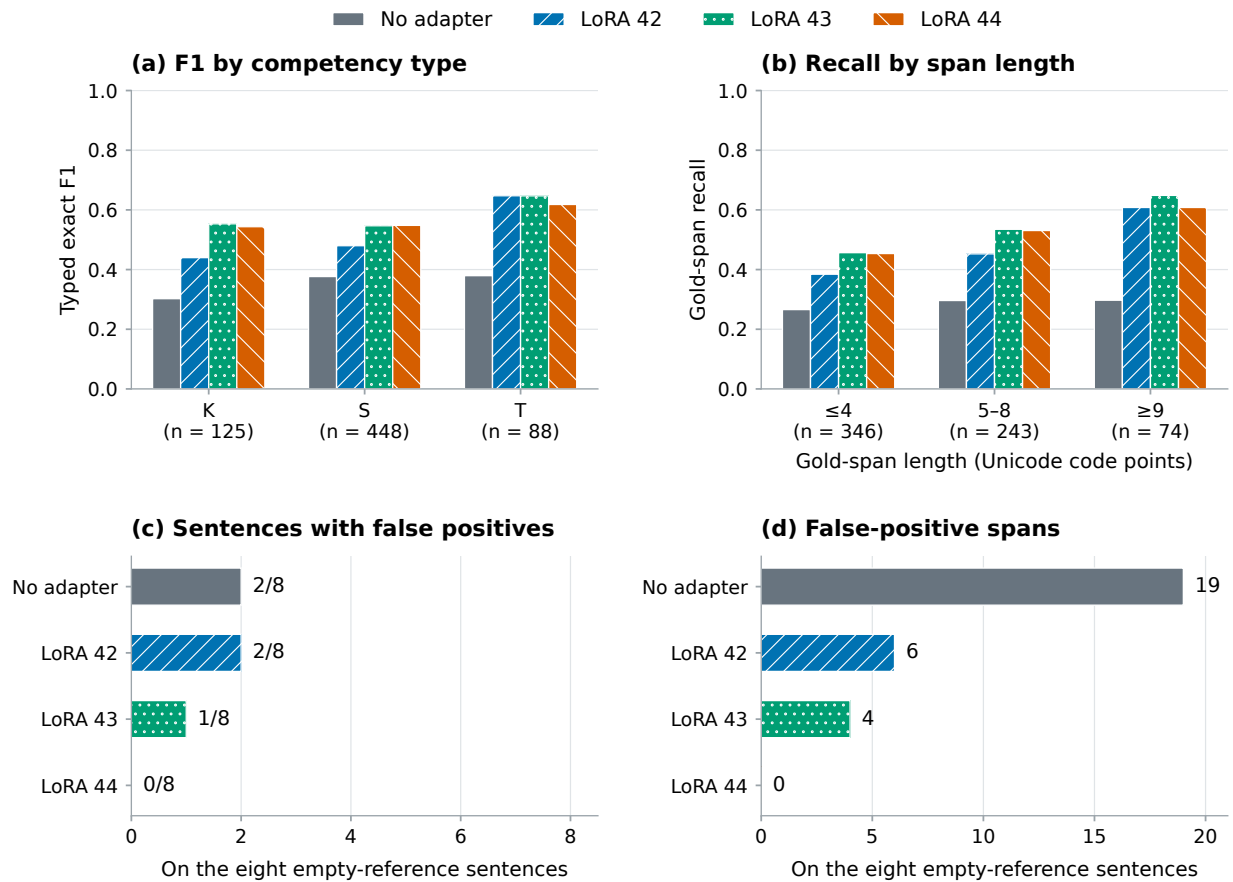


Figure 4. Qwen diagnostics on the same 150 reference sentences. (a) Typed exact F1 by competency type. (b) Recall by gold-span length; n counts gold spans. (c–d) False positives on eight empty-reference sentences, counted by sentence and span. Colours and patterns distinguish the runs.

Diagnostics reuse the same 150 reference sentences and each run's frozen accepted spans. Typed exact matching is completed on each full sentence before aggregation by gold-span length. Bins (≤ 4 , 5–8, ≥ 9 Unicode code points) are fixed in advance; they contain 346/243/74 gold spans in 102/110/56 sentences, respectively. A sentence may contribute gold spans to multiple bins. Type-specific unmatched predictions count as FP even in empty-reference sentences. Each seed is evaluated separately on the same test set.

On the eight empty-reference sentences, accepted false-positive spans decrease from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44; the corresponding affected-sentence counts are 2, 2, 1, and 0.

L has two gold spans: the no-adapter run recovers 0/2 and each LoRA run 2/2, with one extra L prediction in every run. Complete TP/FP/FN, legal-empty-output counts, and parser-rejection diagnostics are retained in the diagnostic records.

Exploratory Expanded-Silver Experiments

Expanded-pool adaptation. The expanded Qwen runs use seed 42 and the shared prompt, occurrence output, parser, and Qwen training settings in Table 5. Loss applies only to assistant output. Both pooled checkpoints are selected at epoch one using Silver-validation exact F1; decoding is deterministic with a 2,048-token output cap. The complete configurations retain the parameter records. The expanded JobBERT runs retain the domain-adapted encoder and inherited CRF. Its latest run uses 8,842 training and 698 development records, so the separate 350-sentence Silver-test interpretation applies to Qwen, not to that JobBERT run.

Table 16 reports the two pooled Qwen runs. The latest Codex variant attains typed exact micro-F1 of 0.809 against its held-out Silver targets and 0.585 against Gold150. The Silver-test score measures agreement with held-out labels generated under the training annotation protocol; the Gold150 score measures agreement with the selected human reference.

Table 16. Expanded-pool Qwen results (seed 42). All scores are typed exact micro-F1 except the final relaxed-F1 column. Gold150 is used only for the additional human-reference evaluation.

Pool variant	Silver validation	Silver test	Gold150 exact	Gold150 relaxed
Proxy: 8,943 / 351 / 352	0.783	0.759	0.597	0.713
Codex: 8,842 / 348 / 350	0.833	0.809	0.585	0.703

Counts in the first column are train / validation / test sentences. Both variants combine earlier B2 and public-prompt annotations; their names identify the annotation channel for the replaced 2,500-sentence subset. They contain 9,646 and 9,540 retained sentences, respectively. Both checkpoints were selected at epoch one with patience two. Differences are computed before rounding: the proxy-minus-Codex Gold150 difference rounds to 0.011 even though subtracting the displayed scores gives 0.012.

The Codex variant has higher agreement with its own Silver test labels but an approximately 0.011 lower exact F1 on Gold150 than the proxy variant. The channel replacement also changes record inclusion: 2,353 of the 2,500 Codex annotations are retained, compared with 2,459 proxy annotations. The contrast therefore describes the two resulting training pools rather than an isolated effect of annotation channel.

The expanded JobBERT runs obtain Gold150 exact F1 of 0.242 for the proxy pool and 0.232 for the Codex pool (seed 42); the latter has relaxed F1 of 0.485. A preceding 7,117-record configuration gives 0.237 ± 0.002 across three seeds. These scores are lower than the historical B2 continuation result, but the expanded runs change the annotation protocol, source composition, and development set alongside training size. The latest JobBERT run selects on 698 development records, whereas Qwen uses 348 validation records and retains 350 for Silver testing. Thus these results do not identify an effect of increasing training size alone. They show that the larger supervision pool does not deliver uniform improvements across the recorded baselines; the matched Qwen comparison remains the direct evidence for adaptation under a fixed inference protocol.

For comparison, the B2 Qwen adapter at seed 42 scores 0.500 on Gold150; its three-seed mean of 0.540 is a different summary. Neither is a matched estimate of an expansion effect because the expanded runs change the supervision pool and selection targets. The cause of the lower expanded JobBERT scores remains unresolved: per-type predictions, truncation/windowing counts, label-version compatibility, and decoding diagnostics must be recovered before attributing the decline to any one factor. Exploratory supervision-subset runs and their selection arrangements are preserved in the supporting experiment records. These single-seed runs are not controlled estimates of training-size or annotation-channel effects.

APPENDIX D. REPRODUCTION AND VERSIONED MATERIALS

The public reproduction guide separates three tasks: inspecting released results, scoring saved predictions, and rerunning training or inference. The paper-to-file index links each table and figure to its supporting materials. The resource naming guide maps manuscript terms to exact file names and model identifiers.

The supporting record index links calculation details, per-run scores, and configurations to their protocols; the earlier process archive preserves exploratory subsets and version history. Qwen adapters and frozen predictions are available in the model-reproduction archive identified in Data Availability. These records support traceability but do not supply all training inputs or establish full reproduction. Table 17 distinguishes access and verification levels.

Collection and Preparation Records

Source names in the collection records include MacroData (马克数据网), operated by 重庆马禾锐信息科技有限公司, and the Recruitment Dataset (招聘数据集, ID 163746) on Alibaba Cloud Tianchi.

The available public-institution records retain announcement titles, source names, URLs, listed publication dates, and geographic metadata where available. Listed publication dates are not collection dates. A complete source-by-stage census of advertisement counts, crawl windows, exclusions, and sentence-to-document mappings has not been recovered.

Audit of Frozen Expanded-Silver Splits

A post hoc audit of the complete frozen expanded-Silver files found that their hashes, counts, and identifier order match the archived split manifests. No identifiers or exact/NFC-normalized sentences are shared across partitions or with the 150-sentence human reference. An additional comparison using NFKC normalization, whitespace removal, and case folding identified the same single training-validation pair in both variants, differing only in full-width versus half-width semicolons; both records have empty span annotations. Three additional punctuation-variant pairs occur within each training set. No corresponding normalized-text overlap was found between training or validation and the Silver test partitions, or between any partition and the human reference. The frozen splits and reported results were retained; the audit does not establish that the validation overlap had no effect. These checks do not establish document-level or semantic near-duplicate isolation.

Table 17. Component access and the reproduction level supported by the identified releases. Local possession is distinguished from public access.

Component	Access route / available objects	Scope and limitation
Core v0.1.3	Dataset DOI in Data Availability: corpus, Gold150 (150), B2 (2,150/169), texts and labels	Download the named version; inspect labels and use its splits. It does not contain the later expanded pools.
Qwen reproduction archive	Model DOI: three B2 adapters, selected expanded adapters, frozen predictions, prompts and scorer	Follow its model index for inference or rescoring; obtain base weights separately. Full training inputs are not bundled.
Final expanded pools	9,540 / 9,646 records; complete files verified locally against split manifests	IDs and selected outputs are archived; public complete texts and labels, or an executable controlled-access arrangement, remain unavailable. Not publicly retrainable as a complete study.
QA150 and historical coding	Pseudonymized texts and human layers linked in Data Availability; source hashes and cohort membership	Recalculate agreement without account metadata. The release preserves design distinctions and does not establish final-rule blind reliability.
Three-annotator blinded study	Separate GitHub agreement release: 50 sentences, frozen labels, source IDs, scorer and intervals	Recalculate independent agreement; five + 15 training sentences excluded. This is not a model test set.
Chinese encoder base-lines	Six public Hugging Face checkpoints; audit DOI 10.5281/zenodo.22937245	Frozen predictions, code and configurations support rescoring. Local rescoring verified all six runs; server checkpoint re-inference is separately recorded.
External experiments	Appendix E links public data/model sources and pinned result archives for 104 server-B and 36 server-A training runs	Metrics and available predictions support checks; unresolved provenance and environment gaps prevent a complete training-package claim.

Component-specific reuse conditions apply. Third-party text is not covered by the authors' annotation permission; model/base-model notices apply separately. No guaranteed access-on-request or exact reconstruction from currently live recruitment pages is asserted.

Resource Access and Sampling Scope

The September 21 audit verified the complete expanded files against their recorded hashes. The files remain author-held; these checks establish file identity, but neither public access nor successful retraining.

Expansion sampling. Preparation retained 2,064 valid unique sentences from the original supervision files and selected 7,936 new candidates: 3,174 cloud, 2,381 public-sector, and 2,381 listed-company sentences. Selection prioritized requirement-related keywords within each source and used fixed hash ranking with seed 20260913. Processing removed HTML and whitespace, split advertisements into sentence units, and kept normalized lengths of 8–400 characters. Matching sentences were excluded against existing corpus splits, supervision, human-review sets, and the available 3.2-million-sentence dump used for the paper's domain pretraining. The compari-

son key applies NFC, removes whitespace, and trims specified edge punctuation. Annotations use the prepared strings. These checks concern sentence matches under that key and do not establish advertisement-level or semantic near-duplicate isolation. The preparation and audit records document the comparison-key function and checked pretraining dumps. The paper-main dump contains 3.2 million lines, but its content hash was not retained. Two other local dumps, not used for exclusion, contain 143 and 4,919 selected-sentence matches. The checks therefore do not establish exclusion from every possible pretraining source.

Table 18. Recovered sampling-stage counts and missing census information. These are sentence counts, not advertisement counts or mutually exclusive industry totals.

Stage	Available counts	Sampling frame / unavailable detail
Original four collections	22,840; split 17,460/2,143/3,237	AI, Cloud, public-sector, graduate recruitment. Per-source ad counts, time windows, attrition and complete document mapping not recovered.
Targeted study lineage	980 conflict-origin + 1,621 other records	150 reference + 2,451 Silver-plus. This is a selected lineage, not a probability sample.
New expansion candidates	Cloud 3,174; public-sector 2,381; listed-company 2,381	Requirement-keyword-positive sentences; fixed hash ranking. Earlier B2 contributes 2,064 retained unique sentences.
Replacement channel sub-set	2,500 candidates; proxy retains 2,459; Codex retains 2,353	Respective exclusions: 41 (1.6%) and 147 (5.9%). A reason-coded attrition breakdown is not established.
Additional QA150	60 / 45 / 45 source quotas	Sampled from new Silver; machine suggestions visible. Does not extend the independent test population.

Annotation Channels and Frozen Targets

The expansion used Codex CLI and a third-party API-compatible proxy as distinct annotation channels, not independently verified model identities. Replacing annotations for 2,500 candidate sentences and excluding unresolved records produced pools of 9,646 and 9,540 records. The version and annotation records retain the stage-specific prompts, selection history, and original labels. A documented experience-boundary check concerns a single Silver example; it does not establish corpus-wide compatibility. Applying revised rules requires a separately versioned annotation release.

Historical implementation. For some JobBERT runs, the exact tokenizer revision, special-token handling, truncation or windowing counts, and affected reference-span counts have not been recovered. The maximum input of 256 tokens is not a character limit, and input-length distributions alone cannot quantify truncation. The reported reference scores retain the full target sets. Per-seed checkpoint-selection records are stored with the available implementation history, including the inherited checkpoint’s unresolved selection history.

APPENDIX E. EXTERNAL BENCHMARK DETAILS

Original-task Reruns and Scoring

We reran BERT (Devlin et al., 2019), SpanBERT (Joshi et al., 2020), JobBERT, and JobSpanBERT (Zhang et al., 2022b) separately for skills and knowledge on SkillSpan’s public HOUSE and TECH subsets. The MaChAmp (van der Goot et al., 2021) seq_bio CRF runs used 20 epochs, development-set checkpoint selection, and five seeds listed in the experiment records. BIG was unavailable; multi-task training was not rerun. Table 20 gives MaChAmp per-site span-F1. The archived pooled scores report BIO rescoring of the same predictions with the correct task column. These scoring procedures yield different results. The native runs contain 3,570 test sentences, compared with 3,569 in the common-protocol release; the one-record difference has not been resolved by matching IDs. For published test references, we use only the original paper’s TEST/STL rows. Its HOUSE/TECH rows report development results. For German public ICT, jobBERT-de used a 20-epoch MaChAmp configuration and ESCOXML-R used its released five-epoch configuration (Zhang et al., 2023). Both used five recorded seeds. Compatibility fields were added for newer MaChAmp without changing the learning rate or epoch limit; the exact changes are archived. The runs therefore use the public releases with compatibility changes, rather than recreating every original environment. They contain 2,557 test sentences, whereas the ESCOXML-R paper reports 2,943. They cover ICT extraction, not the complete German EDU/EXP/LNG pipeline (Gnehm et al., 2022).

Data and Implementation Sources

We obtained the common-protocol data from the public Hugging Face releases of SkillSpan, Kompetenzer, Gnehm ICT, Green, Sayfullina, and FIJO. The corresponding papers are cited in the evaluation protocol; preprocessing and splits in these releases may differ from the original studies. Code is available from the SkillSpan, Kompetenzer, ESCOXML-R, and MaChAmp repositories. The NNOSE repository supplies the retrieval-augmented implementation used in the subsequent A100 experiments. Our linear-head protocol and compatibility changes are described above. Recorded revisions are listed in the experiment records below.

Uncertainty and Experiment Records

Table 19 gives the paired sentence-bootstrap intervals. Sayfullina’s interval includes zero, although its lower endpoint is close to zero. Because the intervals are pointwise, exclusions of zero do not constitute significance tests adjusted for multiple comparisons. The intervals capture sentence sampling uncertainty conditional on the trained seeds; seed SD describes variation across runs. Neither accounts for dependence between sentences from the same advertisement. The versioned experiment records link the server-B common-protocol, Chinese-transfer and native-task runs, and the subsequent server-A Kompetenzer and NNOSE runs. They retain all seeds, model revisions, selected epochs, per-seed scores, and the available configuration and prediction hashes. Some dataset and executed-code identifiers remain unresolved. External-task checks reconciled per-seed summaries with the ledgers; they did not rerun training or independently rescore predictions. For the Chinese encoders, local cnss-lsht-1.2.0 rescoring reproduced all six runs’ exact, relaxed, boundary-only, and per-type scores. Development-selected epochs and source offsets were also checked. Server-reported checkpoint re-inference matched archived tags on all 150 sentences per run; this was not a training rerun. The pinned

848 audit archive and verification record preserve code, configurations, predictions, and the detailed
 849 checks.

Table 19. Paired differences under the common protocol (ESCOXLM-R minus XLM-R).

Dataset	Stream	Mean difference	95% interval
SkillSpan	Skill	+0.010	[−0.003, 0.023]
SkillSpan	Knowledge	−0.027	[−0.041, −0.013]
Kompetencer	Skill	−0.018	[−0.063, 0.022]
Kompetencer	Knowledge	−0.025	[−0.102, 0.056]
Gnehm ICT	ICT	−0.023	[−0.034, −0.013]
Green	Native types	−0.001	[−0.021, 0.017]
Sayfullina	Skill	+0.006	[−1.47 × 10 ^{−5} , 0.011]
FIJO	Native types	−0.004	[−0.043, 0.038]

Pointwise sentence-bootstrap intervals (2,000 resamples; seed 20260921), conditional on seeds 42/43/44. No advertisement clustering or multiple-comparison adjustment. Values are rounded after computing differences and intervals. Three decimal places are used except for Sayfullina’s near-zero lower endpoint, shown with three significant digits in scientific notation to preserve its negative sign. Its interval includes zero.

Table 20. SkillSpan: published single-task TEST references and public-subset MaChAmp reruns (0–1; five seeds).

Model	Task	Published TEST	Rerun HOUSE	Rerun TECH
BERT	Skills	0.543 ± 0.007	0.496 ± 0.013	0.514 ± 0.012
BERT	Knowledge	0.624 ± 0.004	0.550 ± 0.004	0.700 ± 0.008
SpanBERT	Skills	—	0.516 ± 0.014	0.526 ± 0.014
SpanBERT	Knowledge	—	0.536 ± 0.017	0.691 ± 0.015
JobBERT	Skills	0.561 ± 0.005	0.534 ± 0.014	0.538 ± 0.013
JobBERT	Knowledge	0.639 ± 0.003	0.577 ± 0.004	0.708 ± 0.005
JobSpanBERT	Skills	0.566 ± 0.008	0.543 ± 0.013	0.519 ± 0.010
JobSpanBERT	Knowledge	0.611 ± 0.010	0.546 ± 0.018	0.687 ± 0.014

Published values are the TEST/STL results in Table 5 of Zhang et al. (2022b), converted from percent and rounded to three decimal places; their HOUSE/TECH rows report development results. Our reruns use MaChAmp per-site span-F1 on public HOUSE+TECH, excluding BIG. The evaluation samples differ, so score differences cannot be interpreted as matched replication gaps. The source table gives no SpanBERT TEST score. Multi-task training was not rerun.

Table 21. German public ICT extraction: native-task recipes (0–1; five seeds).

Task/model	Recipe	Published	Rerun
Public ICT: jobBERT-de	20-epoch MaChAmp	—	0.885 ± 0.010
Public ICT: ESCOXLM-R	5-epoch released recipe	0.884 ± 0.005	0.891 ± 0.005

Reruns use 2,557 test sentences. The published ESCOXLM-R score comes from Table 2 of Zhang et al. (2023), whose Table 1 reports 2,943 test sentences. These different releases prevent a matched comparison with the published score. No corresponding published jobBERT-de score is supplied. Epoch budgets differ between the two models here, unlike in the common-protocol comparison.

Kompetencer Classification and NNOSE Reruns

Kompetencer fine-grained classification (Zhang et al., 2022a) and NNOSE (Zhang et al., 2024) were run on an NVIDIA A100 (80 GB). Software versions and compatibility records are provided in the native-task configurations. These experiments retain their task-specific metrics and are separate from the common linear-head protocol.

For Kompetencer, MaChAmp used a 20-epoch budget and the five seeds recorded for the SkillSpan native runs. Training used Danish (DA), English (EN), or both (EN+DA), with 138 Danish and 9,472 English training records. All conditions selected checkpoints on the 1,577-record English development set. Table 22 reports unweighted macro-F1 for classification of supplied spans. DaJobBERT resolved to an uncased checkpoint, and the metric differs from the original weighted score; neither detail supports a matched replication claim.

NNOSE used JobBERT and JobBERTa with seed 113412 and development-based early stopping. Each datastore contained only training examples from the evaluated dataset. Neighbour count k , interpolation weight λ , and temperature T were selected on development data and frozen before testing. Table 23 compares sequeval span-F1 from the linear head and retrieval-augmented predictions of the same checkpoint. These linear-head results belong to the NNOSE pipeline, not the common-protocol XLM-R comparison. The all-dataset datastore was not run, and the single-seed results do not establish a general retrieval benefit.

Table 22. Kompetencer fine-grained classification: macro-F1 across five seeds (0–1).

Encoder	Training	Danish test	English test
DaJobBERT	DA	0.182 ± 0.034	0.064 ± 0.012
DaJobBERT	EN	0.232 ± 0.028	0.539 ± 0.010
DaJobBERT	EN+DA	0.340 ± 0.017	0.529 ± 0.008
RemBERT	DA	0.050 ± 0.072	0.034 ± 0.041
RemBERT	EN	0.261 ± 0.019	0.552 ± 0.017
RemBERT	EN+DA	0.316 ± 0.019	0.554 ± 0.014

Mean $\pm$ sample SD; Danish/English test sizes are 784/1,578 classified spans. All conditions use English development data. EN training with Danish testing is the EN-to-DA transfer condition. DaJobBERT resolved to an uncased checkpoint despite the requested cased identifier. Scores are unweighted classification macro-F1, not span-F1 or the source plot’s weighted metric. No matched published-score comparison is claimed.

Table 23. NNOSE with an in-dataset training datastore: test span-F1 (0–1).

Dataset	Encoder	Test n	Linear	kNN
SkillSpan	JobBERT	3,569	0.613	0.612
SkillSpan	JobBERTa	3,569	0.631	0.625
Green	JobBERT	335	0.491	0.500
Green	JobBERTa	335	0.466	0.485
Sayfullina	JobBERT	1,851	0.897	0.896
Sayfullina	JobBERTa	1,851	0.915	0.912

Single seed 113412; no SD is estimated. Selected epochs and development-selected retrieval settings are preserved in the complete run table. Both columns use the same trained checkpoint within each row. The datastore excludes development and test examples. SkillSpan uses public HOUSE+TECH. The all-dataset datastore was not evaluated, and alignment with an original-paper result cell has not been established.

Combined English/Danish training improved Danish macro-F1 for both encoders; RemBERT's Danish-only score varied across seeds (0.050 ± 0.072). English-based selection and the unweighted metric prevent a matched published-score comparison. NNOSE improved Green but slightly reduced SkillSpan and Sayfullina F1 in these single-seed runs, offering no consistent retrieval benefit. The coverage record identifies uncompleted configurations; no missing result is assigned a zero score. These task-specific comparisons broaden method coverage but do not replace an independent Chinese human test.

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COMPETING INTERESTS

The authors declare that they have no competing interests.

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REFERENCES

- Artstein, R. and Poesio, M. (2008). Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.
- Beauchemin, D., Laumonier, J., Le Ster, Y., and Yassine, M. (2022). FIJO: a French Insurance Soft Skill Detection Dataset. In *Proceedings of the Canadian Conference on Artificial Intelligence*. Canadian Artificial Intelligence Association. Article 2022L34.
- Bender, E. M. and Friedman, B. (2018). Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.
- Bhola, A., Halder, K., Prasad, A., and Kan, M.-Y. (2020). Retrieving skills from job descriptions: A language model based extreme multi-label classification framework. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5832–5842. International Committee on Computational Linguistics.
- Cao, L., Zhang, J., Ge, X., and Chen, J. (2021). Occupational profiling driven by online job advertisements: Taking the data analysis and processing engineering technicians as an example. *PLOS ONE*, 16(6):e0253308.

- Chen, C. and Kong, F. (2021). Enhancing entity boundary detection for better chinese named entity recognition. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 20–25. Association for Computational Linguistics.
- Chen, X., Qin, C., Fang, C., Wang, C., Zhu, C., Zhuang, F., Zhu, H., and Xiong, H. (2024). Job-SDF: A multi-granularity dataset for job skill demand forecasting and benchmarking. In *Advances in Neural Information Processing Systems*, volume 37.
- Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.
- Council of the European Union (2017). Council recommendation of 22 may 2017 on the european qualifications framework for lifelong learning. Official Journal of the European Union, C 189, pp. 15–28. Accessed: 2026-08-28.
- De Santo, A., Malandri, L., Mercorio, F., Mezzanzanica, M., and Nobani, N. (2026). SkiLLens: Recognising and mapping novel skills from millions of job ads across europe using language models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 877–885. Association for Computational Linguistics.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics.
- Ding, B., Qin, C., Liu, L., Chia, Y. K., Li, B., Joty, S., and Bing, L. (2023). Is GPT-3 a good data annotator? In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11173–11195. Association for Computational Linguistics.
- European Commission (2024). ESCO skills pillar: Skills and competences. https://esco.ec.europa.eu/en/classification/skill_main. Version 1.2.0 used to define the annotation inventory; accessed: 2026-08-28.
- Fetahu, B., Kar, S., Chen, Z., Rokhlenko, O., and Malmasi, S. (2023). SemEval-2023 task 2: MultiCoNER II multilingual complex named entity recognition. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval 2023)*, pages 2247–2265. Association for Computational Linguistics.
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., and Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.
- Gnehm, A.-S., Bühlmann, E., Buchs, H., and Clematide, S. (2022). Fine-grained extraction and classification of skill requirements in German-Speaking job ads. In *Proceedings of the Fifth Workshop on Natural Language Processing and Computational Social Science (NLP+CSS)*, pages 14–24. Association for Computational Linguistics.
- Green, T., Maynard, D., and Lin, C. (2022). Development of a benchmark corpus to support entity recognition in job descriptions. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 1201–1208, Marseille, France. European Language Resources Association.

- Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., and Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360. Association for Computational Linguistics.
- Hripcsak, G. and Rothschild, A. S. (2005). Agreement, the F-measure, and reliability in information retrieval. *Journal of the American Medical Informatics Association*, 12(3):296–298.
- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., and Levy, O. (2020). SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics*, 8:64–77.
- Kim, K. W., Kim, J.-D., and Aizawa, A. (2026). Refining and reusing annotation guidelines for LLM annotation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 37951–37964. Association for Computational Linguistics.
- Klie, J.-C., Eckart de Castilho, R., and Gurevych, I. (2024). Analyzing dataset annotation quality management in the wild. *Computational Linguistics*, 50(3):817–866.
- Lafferty, J., McCallum, A., and Pereira, F. C. N. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the Eighteenth International Conference on Machine Learning*, pages 282–289. Morgan Kaufmann.
- le Vrang, M., Papantoniou, A., Pauwels, E., Fannes, P., Vandensteen, D., and De Smedt, J. (2014). ESCO: Boosting job matching in europe with semantic interoperability. *Computer*, 47(10):57–64.
- Li, M., Shi, T., Ziems, C., Kan, M.-Y., Chen, N., Liu, Z., and Yang, D. (2023). Coannotating: Uncertainty-guided work allocation between human and large language models for data annotation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1487–1505. Association for Computational Linguistics.
- Lu, Y., Liu, Q., Dai, D., Xiao, X., Lin, H., Han, X., Sun, L., and Wu, H. (2022). Unified structure generation for universal information extraction. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5755–5772. Association for Computational Linguistics.
- Luo, H., Jin, Y., Wang, Y., Li, X., Shang, T., Liu, X., Chen, R., Wang, K., Salam, H., Wen, Q., and Liu, Z. (2025). DynamicNER: A dynamic, multilingual, and fine-grained dataset for LLM-based named entity recognition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 16511–16535. Association for Computational Linguistics.
- MacroData (n.d.). Macrodata (马克数据网). <https://www.macrodatas.cn/>. Platform operated by 重庆马禾锐信息科技有限公司. Accessed 20 September 2026.
- Magron, A., Dai, A., Zhang, M., Montariol, S., and Bosselut, A. (2024). JobScape: A framework for generating synthetic job postings to enhance skill matching. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 43–58. Association for Computational Linguistics.
- Nguyen, K., Zhang, M., Montariol, S., and Bosselut, A. (2024). Rethinking skill extraction in the job market domain using large language models. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 27–42. Association for Computational Linguistics.
- Peng, N. and Dredze, M. (2015). Named entity recognition for chinese social media with jointly

- trained embeddings. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 548–554.
- Podjanin, A. and Strietska-Ilina, O. (2020). The feasibility of using big data in anticipating and matching skills needs. Technical report, International Labour Organization, Geneva. Accessed: 2025-10-28.
- Sainz, O., García-Ferrero, I., Agerri, R., Lopez de Lacalle, O., Rigau, G., and Agirre, E. (2024). GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations*, volume 2024, pages 47083–47107.
- Sayfullina, L., Malmi, E., and Kannala, J. (2018). Learning representations for soft skill matching. In *Analysis of Images, Social Networks and Texts*, volume 11179 of *Lecture Notes in Computer Science*, pages 141–152. Springer.
- Senger, E., Zhang, M., van der Goot, R., and Plank, B. (2024). Deep learning-based computational job market analysis: A survey on skill extraction and classification from job postings. In *Proceedings of the Workshop on Natural Language Processing for Human Resources (NLP4HR)*, pages 1–15, St. Julian’s, Malta. Association for Computational Linguistics.
- Tianchi (n.d.). 招聘数据集. <https://tianchi.aliyun.com/dataset/163746/>. Recruitment dataset (English translation of title); dataset ID 163746. Accessed 20 September 2026.
- Ul Haq, M. U., Rigoni, D., and Sperduti, A. (2026). A systematic comparison of large language models for data annotation in NER tasks. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 524–548. ELRA Language Resource Association.
- van der Goot, R., Üstün, A., Ramponi, A., Sharaf, I., and Plank, B. (2021). Massive choice, ample tasks (MaChAmp): A toolkit for multi-task learning in NLP. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 176–197. Association for Computational Linguistics.
- Wang, S., Sun, X., Li, X., Ouyang, R., Wu, F., Zhang, T., Li, J., Wang, G., and Guo, C. (2025). GPT-NER: Named entity recognition via large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4257–4275. Association for Computational Linguistics.
- Xu, L., Dong, Q., Liao, Y., Yu, C., Tian, Y., Liu, W., Li, L., Liu, C., and Zhang, X. (2020). CLUENER2020: Fine-grained named entity recognition for Chinese. arXiv preprint arXiv:2001.04351[cs.CL].
- Zaratiana, U., Tomeh, N., Holat, P., and Charnois, T. (2024). GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5364–5376, Mexico City, Mexico. Association for Computational Linguistics.
- Zhang, M., Jensen, K. N., and Plank, B. (2022a). Kompetencer: Fine-grained skill classification in danish job postings via distant supervision and transfer learning. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 436–447, Marseille, France. European Language Resources Association.
- Zhang, M., Jensen, K. N., Sonniks, S. D., and Plank, B. (2022b). SkillSpan: Hard and soft skill extraction from english job postings. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4962–4984, Seattle, United States. Association for Computational Linguistics.
- Zhang, M., Jensen, K. N., van der Goot, R., and Plank, B. (2022c). Skill extraction from job post-

ings using weak supervision. In *Proceedings of the 2nd Workshop on Recommender Systems for Human Resources*, volume 3218 of *CEUR Workshop Proceedings*.

Zhang, M., van der Goot, R., Kan, M.-Y., and Plank, B. (2024). NNOSE: Nearest neighbor occupational skill extraction. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 589–608. Association for Computational Linguistics.

Zhang, M., van der Goot, R., and Plank, B. (2023). ESCOXML-R: Multilingual taxonomy-driven pre-training for the job market domain. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11871–11890. Association for Computational Linguistics.

Zhong, J., Xuan, Z., Wang, K., and Cheng, Z. (2023). A BERT-Span model for Chinese named entity recognition in rehabilitation medicine. *PeerJ Computer Science*, 9:e1535.

Zhou, W., Zhang, S., Gu, Y., Chen, M., and Poon, H. (2024). UniversalNER: Targeted distillation from large language models for open named entity recognition. In *The Twelfth International Conference on Learning Representations*.